G.C.d. (Greatest Common divisor)

Defn. The G.c.d of two integers a and b, that are not both zero, is the largest integer which divides both a and b.

The G.c.d. of a and b is denoted by (a, b).

Eg: The common divisors of 24 and 84 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6$ and ± 12. Hence G.c.d of 24, 84 is 12

i.e $(24, 84) = 12$

$(15, 81) = 3$, $(100, 5) = 5$, $(17, 25) = 1$, $(0, 44) = 44$, $(-6, -15) = 3$ and $(-17, 289) = 17$.

Relatively prime numbers:

The integers a and b are called relatively prime if a and b have greast common divisor $(a, b) = 1$.

Eg: Since $(25, 42) = 1$; 25 and 42 are relatively prime

Note: $(a, b) = (|a|, |b|)$ where $|a|$ denotes the absolute realue of a.

$$|a| = \begin{cases} a ; a > 0 \\ -a \quad a \leq 0 \end{cases}$$

Theorem: let a, b and c be integers with $(a, b) = d$ then

(i) $(a/d, b/d) = 1$ ii) $(a + cb, b) = (a, b)$.

Proof: i) let a and b be integers with $(a, b) = d$.

To prove: $\left(\frac{a}{d}, \frac{b}{d}\right) = 1$.

Assume that e is a positive integer such that $e/\left(\frac{a}{d}\right)$

and $e \mid (b/d)$.

$\Rightarrow \exists$ integers k and l such that $\frac{a}{d} = ke$ and $\frac{b}{d} = le$

$\Rightarrow a = dek$ and $b = del$

$\Rightarrow de \mid a$ and $de \mid b$ i.e de is a common divisor of a & b

since $(a, b) = d$

i.e d is the greast common divisor of a and b

$\therefore$ ~~de~~ e must be 1 i.e $e = 1$

$\therefore \left(\frac{a}{d}, \frac{b}{d}\right) = 1$

ii) let a, b and c be integers.

To prove : $(a+cb, b) = (a, b)$

It is enough to prove that the common divisors of a and b exactly the same as the common divisors of $a+cb$, and b

let e be a common divisor of a and b

$\Rightarrow e \mid a$ and $e \mid b$

$\Rightarrow e \mid a+cb$

$\Rightarrow e$ is a common divisor of $a+cb$ and b.

If f is a common divisor of $a+cb$, and b

$\Rightarrow f \mid a+cb$ and $f \mid b$

$f \mid b \Rightarrow f \mid cb$

$\Rightarrow f \mid a+cb-cb$

$\Rightarrow f \mid a$

$\Rightarrow$ d is a common divisor of a and b. (2)

Hence $(a+cb, b) = (a, b)$.

Defn. Linear Combination of a and b:

If a and b are integers, then a linear combination of a and b is a sum of the form $ma+nb$ where both m and n are integers.

$1\cdot 9 + (-1)\,15 = -6$ $\quad$ $0\cdot 9 + 0\cdot 15 = 0$

$(-2)\,9 + 1\cdot 15 = -3$ $\quad$ $2\cdot 9 + (-1)\,15 = 3$

$(+1)\,9 + 1\cdot 15 = 6$ and so on.

set of all linear combinations of 9 & 15 is $\{\ldots, -12, -9, -6, -3, 0, 3, 6, 9, 12, \ldots\}$.

Theorem (2). The greatest common divisor of the integers a and b, that are not both zero, is the least positive integer that is a linear combination of a and b.

Proof: ~~Let d be the least positive integer which~~ (OR)

If $(a, b) = d$ then there exists $x, y \in \mathbb{Z}$ such that $d = ax+by$ ~~which is~~ and d is the least positive value of $ax+by$ where x, y range over all integers.

Proof: Let $S = \{ax+by \mid ax+by > 0 \text{ and } x, y \in \mathbb{Z}\}$

Since $a^2 = a\cdot a + b\cdot 0 > 0$ for $a, 0 \in \mathbb{Z}$,

$\Rightarrow a^2 \in S$

$\therefore S \neq \Phi \subseteq N$.

$\therefore$ By well-ordering principle (Every non-empty set of positive integers has a least member),

S has a least number, say s.

$\therefore s \in = ax_1 + by_1$ for $x_1, y_1 \in \mathbb{Z}$

We prove that: $s = (a, b) = d$.

Since $d = (a,b) \Longrightarrow d/a$ and d/b

$\Longrightarrow d/ax_1 + by_1$

$\Longrightarrow d/s \longrightarrow (1)$

If $s \nmid a$, then by division algorithm, we have

$a = qs + r$ where $0 < r < s$

$\Longrightarrow r = a - qs = a - q(ax_1 + by_1) = a(1 - qx_1) + b(-qy_1)$

$\Longrightarrow r = ax_2 + by_2$

where $x_2 = 1 - qx_1$ and $y_2 = -qy_1$

$\therefore r \in S$

we have $r \in S$ and $r < s$

which is a contradiction as s is the least number

$\therefore s/a$

Why we can say that s/b

$\therefore s$ is a common divisor of a and b

Since $d = (a,b) \Rightarrow s/d \longrightarrow (2)$

$\therefore$ from (1), (2)

$s = d$

Hence $d = ax_1 + by_1$ for $x_1, y_1 \in \mathbb{Z}$

Since d is the least member of S, d is the least positive value of $ax + by$ where $x, y \in \mathbb{Z}$.

If a and b are integers not both 0 then a positive integer d is the g.c.d (a,b) if and only if i) $d|a$ & $d|b$ and ii) if c is an integer with $c|a$ and $c|b$ then $c|d$.

Note: The integers a and b are relatively prime integers if and only if there are integers m and n such that $ma + nb = 1$.

If a & b are positive integers then the set of linear combinations of a and b is the set of integer multiples of (a,b)

Bezout's theorem: If a and b are integers, then there are integers m & n such that $ma + nb = (a,b)$.

Here $ma + nb = (a,b)$ is Bezout's identity and m, n are called Bezout coefficients or Bezout numbers of the pair of integers a & b.

Defn: Let $a_1, a_2, a_3 \ldots a_n$ be integers, that are not all zero, the g.c.d. of these integers is the largest integer which is a divisor of all of the integers in the set.

The g.c.d. of $a_1, a_2, \ldots a_n$ is denoted by $(a_1, a_2, a_3, \ldots a_n)$.

eg: $(12, 18, 30) = 6$, $(10, 15, 25) = 5$

Lemma:

Theorem: If $a_1, a_2, \ldots a_n$ are integers, that are not all zero, then

$(a_1, a_2, \ldots a_n) = (a_1, a_2, \ldots, (a_{n-1}, a_n))$.

Proof: Any common divisor of n integers $a_1, a_2, a_3, \ldots a_n$ is also a divisor of a_{n-1} and a_n

$\therefore$ It is also a divisor of (a_{n-1}, a_n)

Any common divisor of the $n-2$ integers $a_1, a_2, \ldots a_{n-2}$ and (a_{n-1}, a_n) must be a common divisor of all n integers.

Since it divides (a_{n-1}, a_n), it must divide both a_{n-1} and a_n.

$\therefore$ Any common divisor of n integers $a_1, a_2, \ldots a_n$ is also a common divisor of $n-2$ integers $a_1, a_2, \ldots a_{n-2}$ and (a_{n-1}, a_n).

$\therefore$ $(a_1, a_2, a_3, \ldots a_n) = (a_1, a_2, a_3, \ldots (a_{n-1}, a_n))$

Ex: $(105, 140, 350) = (105, (140, 350)) = (105, 70) = 35$

$(15, 21, 35) = (15, (21, 35)) = (15, 7) = 1$

Defn: Mutually relatively prime:

The integers $a_1, a_2, \ldots a_n$ are mutually relatively prime if $(a_1, a_2, a_3, \ldots a_n) = 1$.

Pairwise relatively prime:

The integers $a_1, a_2, \ldots a_n$ are pairwise relatively prime if each pair of integers from the set is relatively prime.

Lemma (2): If c and d are integers and $c = dq + r$ where q and r are integers then $(c, d) = (d, r)$.

Proof: If an integer e divides both c and d since $r = c - dq$

we have $e \mid c - dq \Rightarrow e \mid r$

$\therefore$ $e \mid r$ and $e \mid d$ $\longrightarrow$ (1)

If $f \mid d$ and $f \mid r \Rightarrow f \mid dq + r$ for some $q \in \mathbb{Z}$

$\Rightarrow f \mid c$ since $c = dq + r$

i.e $f \mid c$ and $f \mid d$ $\longrightarrow$ (2)

i.e the common divisors of c and d are the same as the common divisors of d and r and vice versa.

$\therefore (c, d) = (d, r)$

The Euclidean Agorithm:

Let $r_0 = a$ and $r_1 = b$ be nonnegative integers with $b \neq 0$. If the division algorithm is successively applied to obtain

$r_j = r_{j+1} \cdot q_{j+1} + r_{j+2}$ with $0 < r_{j+2} < r_{j+1}$ for $j = 0, 1, 2, \ldots, n-2$ and $r_n \neq 0$, $a = bq_1 + r_2$ $0 < r_2 < b$ then $(a, b) = r_n$, the last nonzero remainder.

Proof: Let $r_0 = a$ and $r_1 = b$ be positive integers with $a \geq b$.

By successively applying the division algorithm, we find that

$r_0 = r_1 q_1 + r_2$ $\quad 0 \leq r_2 < r_1$

$r_1 = r_2 q_2 + r_3$ $\quad 0 \leq r_3 < r_2$

$r_2 = r_3 q_3 + r_4$ $\quad 0 \leq r_4 < r_3$,

$r_{n-3} = r_{n-2} q_{n-2} + r_{n-1}$ $\quad 0 \le r_{n-1} < r_{n-2}$ (4)

$r_{n-2} = r_{n-1} q_{n-1} + r_n$ $\quad 0 \le r_n < r_{n-1}$

$r_{n-1} = r_n q_n$

We can assume that we eventually obtain a remainder of zero since the sequence remainders $a = r_0 > r_1 > r_2 > r_3 > \cdots \geq 0$ cannot contain more than a terms.

∴ By previous lemma, we have (r_{n-2}, r_{n-1})

$(r_0, r_1) = (r_1, r_2) = (r_2, r_3) = (r_3, r_4) = \cdots = (r_{n-3}, r_{n-2})$

$= (r_{n-2}, r_{n-1}) = (r_{n-1}, r_n) = (r_n, 0) = r_n$

∴ $(a, b) = r_n$, the last nonzero remainder.

Fibonacci numbers:

The Fibonacci numbers $u_1, u_2, u_3, \ldots$ are defined recursively by the equations $u_1 = u_2 = 1$ and $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$.

Theorem (3): Let n be a positive integer and let $\alpha = \left(\frac{1+\sqrt{5}}{2}\right)$. Then $u_n > \alpha^{n-2}$ ∀ for $n \geq 3$.

Proof:

we have $\alpha = \frac{1+\sqrt{5}}{2} < 2 = u_3$

$\Rightarrow \alpha < u_3$

$\Rightarrow u_3 > \alpha^1$

i.e the theorem is true for $n = 3$.

Assume that the theorem is true for all integers k with $n \geq k$

i.e we have $u_k > \alpha^{k-2}$ is true for $n \geq k$.

Since $\alpha = \frac{1+\sqrt{5}}{2}$ is a solution of $x^2 - x - 1 = 0$, we have

$\alpha^2 = \alpha + 1$.

Hence $\alpha^{k-1} = \alpha^2 \cdot \alpha^{k-3} = (\alpha+1)\alpha^{k-3}$ $(\because \alpha^2 = \alpha+1)$

$\alpha^{n-1} = \alpha^2 \cdot \alpha^{n-3} = (\alpha+1)\alpha^{n-3}$

$\Longrightarrow \alpha^{k-1} = \alpha^{k-2} + \alpha^{k-3}$

$\alpha^{n-1} = \alpha^{n-2} + \alpha^{n-3}$

By the induction hypothesis, we have

$\alpha^{n-2} < u_n$ $\alpha^{k-2} < u_k$ and $\alpha^{k-3} < u_{k-1}$ $\alpha^{n-3} < u_{n-1}$

$\therefore \alpha^{k-1} = \alpha^{k-2} + \alpha^{k-3} < u_k + u_{k-1} = u_{n+1}$

$\Longrightarrow \alpha^{k-1} < u_{k+1}$ $u_{n+1} > \alpha^{n-1}$ for $n \geq k+1$.

ie The result is true $n \geq k+1$.

$\therefore$ By the principle of mathematical induction, the result $u_n > \alpha^{n-2}$ is true for $n \geq 3$.

Theorem ④: let u_{n+1} and u_{n+2} be successive terms of the Fibonacci Sequence. Then the Euclidean algorithm takes exactly n divisions to show that $(u_{n+1}, u_{n+2}) = 1$.

Proof: Applying the Euclidean algorithm, and using the defining relation for the Fibonacci numbers, $u_j = u_{j-1} + u_{j-2}$ in each step, we see that

$$u_{n+2} = u_{n+1} \cdot 1 + u_n,$$
$$u_{n+1} = u_n \cdot 1 + u_{n-1},$$
$$\vdots$$
$$u_4 = u_3 \cdot 1 + u_2,$$
$$u_3 = u_2 \cdot 2 +$$

$\therefore (u_{n+2}, u_{n+1}) = (u_{n+1}, u_n) = \cdots = (u_4, u_3) = (u_3, u_2) = (u_2, 0) = u_2$

$\Longrightarrow (u_{n+2}, u_{n+1}) = u_2 = 1$.

Hence the Euclidean algorithm takes exactly n divisions, to show that $(u_{n+2}, u_{n+1}) = 1$.. (5)

Lame's theorem :

The number of divisions needed to find the greatest common divisor of two positive integers using the Euclidean algorithm does not exceed five times the number of decimal digits in the smaller of the two integers.

Proof: When we apply the Euclidean algorithm to find the g.c.d. of $a = r_0$ and $b = r_1$ with $a > b$, we obtain the following sequence equations :

$$a = bq_1 + r_2$$

$$\Rightarrow r_0 = r_1 q_1 + r_2 \quad \text{where } 0 \le r_2 < r_1$$

$$r_1 = r_2 q_2 + r_3 \qquad 0 \le r_3 < r_2$$

$$r_2 = r_3 q_3 + r_4 \qquad 0 \le r_4 < r_3$$

$$\vdots \qquad \vdots$$

$$r_{n-2} = r_{n-1} q_{n-1} + r_n \quad , \quad 0 \le r_n < r_{n-1}$$

$$r_{n-1} = r_n q_n .$$

We have used n divisions.

Since $r_n < r_{n-1}$, $q_n \ge 2$

and each of the quotients $q_1, q_2, q_3, \ldots q_{n-1}$ is greater than or equal to 1 i.e $q_i \ge 1$ for $i = 1, 2, \ldots n-1$.

$\therefore \quad r_n \ge 1 = u_2$

$q_{n-1} \ge 1$

$r_{n-1} q_{n-1} \ge r_{n-1}$

$r_{n-1} q_{n-1} + r_n > r_{n-1}$

$r_{n-1} > r_{n-1} + r_n$

$r_{n-1} \ge 2 r_n \ge 2 u_2 = u_3$ → $q_n \ge 2$, $r_n q_n \ge 2 r_n$, $r_{n-1} \ge 2 r_n$

$r_{n-2} \ge r_{n-1} + r_n \ge u_3 + u_2 = u_4$

$r_{n-3} \geq r_{n-2} + r_{n-1} \geq u_4 + u_3 = u_5$

$r_2 \geq r_3 + r_4 \geq u_{n-1} + u_{n-2} = u_n$

$\Rightarrow b = r_1 \geq r_2 + r_3 \geq u_n + u_{n-1} = u_{n+1}$

$\Rightarrow$ we got finally $b \geq u_{n+1}$

$u_n > \alpha^{n-2} \; \forall n \geq 3$

but $u_{n+1} > \alpha^{n-1}$ for $n > 2$ where $\alpha = \frac{1+\sqrt{5}}{2}$ (from theorem ②)

$\Rightarrow b > \alpha^{n-1}$ $\qquad \alpha = \frac{1+\sqrt{5}}{2} \quad \log_{10} \alpha > \frac{1}{5}$

$\log_{10} b > (n-1) \log_{10} \alpha > \frac{(n-1)}{5}$

$\Rightarrow (n-1) < 5 \cdot \log_{10} b$

Let b have k decimal digits, so that $b < 10^k$ and $\log_{10} b < k$

$\therefore n-1 < 5 \log_{10} b < 5k$

$\Rightarrow n-1 < 5k$

$\Rightarrow n < 5k+1$

$\rightarrow n \leq 5k$ ($\because$ since k is an integer)

(55, 34)

$5 \times 9 = 10$

$55 = 1 \times 34 + 21 = 1$

$34 = 1 \times 21 + 13 = 2$

$21 = 1 \times 13 + 8 = 3$

$13 = 1 \times 8 + 5 = 4$

$8 = 1 \times 5 + 3 = 5$

$5 = 1 \times 3 + 2 = 6$

$3 = 1 \times 2 + 1 = 7$

$2 = 1 \times 1 + 0 = 8$

$(34, 55) = 1$

8 divisions < 10

5×10

$10 = 2$

Problem: Expressing GCD as linear combinations

If $d = (826, 1890)$ using division algorithm compute d and then express as a linear combination of 826, 1890

(or)

Find $x, y \in \mathbb{Z}$ such that $826x + 1890y = (826, 1890)$

Sol.: By Euclid algorithm:

$1890 = 826(2) + 238 \quad ; \; r_2 = 238$

$826 = 238(3) + 112 \quad ; \; r_3 = 112$

$r_2 = 14(8) + 0 \quad ; \quad r_3 = 0$

$\therefore r_4 = 14$ is the last non-zero remainder

$\therefore d = $ G.c.d of $826, 1890 = 14.$

Again $(252, 198) = 18$

$$d = 14 = 238 - 112(2)$$

$$= 238 + (-2)(826 - 3(238))$$

$$= 238 + (-2)\,826 + 6(238)$$

$$= 7(238) + (-2)826$$

$$= [1890 - 2(826)]7 + (-2)826$$

$$= 7(1890) - 14(826) + (-2)826$$

$$= 7(1890) + (-16)\,826$$

$$= (7)\,1890 + (-16)\,826$$

$\therefore d = 14 = 826x + 1890y$ where $x = -16, y = 7.$

7 If $(a,b) = 1$ then show that $(a+b, a-b) = 1$ (or) 2

$a|y \Rightarrow a|z$
$\Rightarrow x|y+z$
$x|y-z$

Let $(a+b, a-b) = d$

$\therefore d|a+b$ and $d|a-b \Longrightarrow d|2a$ and $d|2b$

$\Longrightarrow d$ is a common divisor of $2a, 2b \Longrightarrow d|(2a, 2b)$

wt $(a,b) = 1 \Longrightarrow (2a, 2b) = 2|(a,b) = 2$

$\therefore d|2 \Longrightarrow d = 1$ (or) $d = 2.$

$\therefore (a+b, a-b) = 1$ (or) $2.$

If $(a,b) = d$ then $(ka, kb) = |k|(a,b) = |k|d$ where $k \in \mathbb{Z} \setminus \{0\}$.

Given $(a,b) = d$, $d|a$ and $d|b$ $\quad ka = kdq_1$, $kb = kdq_2$

$\Longrightarrow a = dq_1, \; b = dq_2$

$\exists\, u, v \in \mathbb{Z} \Rightarrow au + bv = d \Rightarrow kau + kbv = kd$ $\quad (ka, kb) = kd = |k|d$ where

$\therefore r_{n-3} = r_{n-2} + r_{n-1} \geq 4b + 4b = 8b$

$r_{n-3} \geq r_{n-2} + \quad + b)$

$\vdots$

$(a, a+b) = l$

$d/a+b \Rightarrow d/(a, a+b)$

$\Rightarrow d/l \longrightarrow ①$

l/a and $l/a+b \Rightarrow l/a+b-a \Rightarrow l/b$

$\therefore l/b$ and $l/a \Rightarrow l/(a,b)$

$\Rightarrow l/d \longrightarrow ②$

$\therefore$ from ①, ② $l = d$

$\therefore (a,b) = (a, a+b)$

If $d = (a,b) = ax+by$ prove that x, y are relatively prime

Soln. Suppose that $(x,y) = d$ l.

Theorem: Let a and b be positive integers. Then $(a,b) = s_n a + t_n b$ for $n = 0,1,2,\ldots$ where s_n and t_n are the nth terms of the sequences defined recursively by $s_0 = 1$, $t_0 = 0$, $s_1 = 0$, $t_1 = 1$ and

$$s_j = s_{j-2} - s_{j-1} \cdot q_{j-1} \quad \text{and} \quad t_j = t_{j-2} - t_{j-1} \cdot q_{j-1} \quad \text{for } j = 2,3,\ldots,n.$$

where the q_j's are the quotients in the divisions of the Euclidean algorithm when it is used to find (a,b).

Proof: We will prove that $r_j = s_j a + t_j b$ for $j = 0,1,2,\ldots n$. $\longrightarrow$ ①

Since $(a,b) = r_n$, once we have established eqn ①, we will know that

$$r_n = (a,b) = s_n a + t_n b.$$

We have $a = r_0 = 1 \cdot a + 0 \cdot b = s_0 a + t_0 b$

i.e the equation ① is valid for $j = 0$.

and $b = r_1 = 0 \cdot a + 1 \cdot b = s_1 a + t_1 b$

i.e the equation ① is valid for $j = 1$.

Assume that eqn ① is valid for $j = 0,1,2,3,\ldots k-1$

i.e $r_j = s_j a + t_j b$ for $j = 1,2,3,\ldots,k-1$.

From the kth step of the euclidean algorithm, we have

$$r_k = r_{k-2} - r_{k-1} \cdot q_{k-1}$$

Now $r_k = (s_{k-2} a + t_{k-2} b) - (s_{k-1} a + t_{k-1} b) q_{k-1}$

$$= (s_{k-2} - s_{k-1} \cdot q_{k-1}) a + (t_{k-2} - t_{k-1} \cdot q_{k-1}) b$$

$$r_k = s_k a + t_k b$$

$\Rightarrow$ The result eqn ① is true $j = k$.

$\therefore r_n = s_n a + t_n b$

$\Rightarrow (a,b) = s_n a + t_n b$

$n-3 \quad n-2 \quad n-1$

Problem: Use the Euclidean algorithm to find the g.c.d of (102, 222) and express the gcd as a linear combination of 102 & 222.

Soln

we have $b = 102 = r_0$, $a = 222 = r_1$

$(102, 222) = r_n = s_n a + t_n b = s_n 222 + t_n 102$

$s_j = s_{j-2} - s_{j-1} q_{j-1}$, $s_0 = 1$, $t_0 = 0$

$s_1 = 0$, $t_1 = 1$

$t_j = t_{j-2} - t_{j-1} \cdot q_{j-1}$

$q_1 = 2$

$s_2 = s_0 - s_1 q_1 = 1 - 0.q_1 = 1$ $\quad t_2 = t_0 - t_1 . q_1 = 0 - 1.q_1 = -q_1$

$t_2 = -2$ $\quad t_2 = -2$.

j	r_j	r_{j+1}	q_{j+1}	r_{j+2}	s_j	t_j
0	222	102	2	18	01	0
1	102	18	5	12	0	01
2	18	12	1	6	1	-2
3	12	6	2		-5	11
4					6	-13

$s_3 = s_1 - s_2 \cdot q_2 = 0 - (1)5 = -5$

$t_3 = t_1 - t_2 \cdot q_2 = 1 + 2(5) = 11$

$s_4 = s_2 - s_3 \cdot q_3 = 1 - (-5).(1) = 1 + 5 = 6$

$t_4 = t_2 - t_3 \cdot q_3 = (-2) - 11(1) = -2 - 11 = -13.$

$r_4 = s_4 a + t_4 b \Rightarrow 6 = 6(222) + (-13).(102)$

Fundamental theorem of Arithmetic:

Every positive integer can be written uniquely as a product of primes, with the prime factors in the product written in order of nondecreasing size.

eg. $240 = 2.2.2.2.3.5 = 2^4.3.5$

$289 = 17.17 = 17^2$

$1001 = 7.11.13.$

Theorem: If $(a,b) = d = 1$ and a/bc then a/c (Gauss lemma)

Proof: $a/bc \Rightarrow bc = aq_1 \;,\; q_1 \in \mathbb{Z}$

$(a,b) = 1 \Rightarrow \exists\; x, y \in \mathbb{Z} \Rightarrow ax + by = 1$

$ax + by = 1$

$\Rightarrow c(ax + by) = c$

$\Rightarrow cax + cby = c$

$\Rightarrow c(ax) + (aq_1)y = c$

$\Rightarrow a(cx + q_1 y) = c$

$\Rightarrow$ ∵ $aq = c$ where $q = cx + q_1 y$

$\Rightarrow a/c$

Euclid's lemma:

If p is a prime and $a, b \in \mathbb{Z}$. then $p/ab \Rightarrow p/a$ (or) p/b

Soln: case ① If p/b then the theorem is proved

case ② If $p \not/ b$ then $(p,b) = 1$ since p is a prime.

$(p,b) = 1 \Rightarrow \exists\; x, y \in \mathbb{Z} \Rightarrow px + by = 1$

$\Rightarrow apx + aby = a \longrightarrow ①$

$p/ab \Rightarrow ab = pq$ where $q \in \mathbb{Z} \longrightarrow ②$

from ①, ②, we have $apx + pq_1 y = a$

$\Rightarrow p(ax + q_1 y) = a$

$\Rightarrow pq' = a$ where $q' = ax + q_1 y$

$\therefore p/a$

Hence $p/ab \Rightarrow$ ~~$p \nmid a$ (or) $p \nmid b$~~ p/a (or) p/b

Note ① If p is a prime and $a_1, a_2, \ldots a_n \in \mathbb{Z}$ then $p/a_1 a_2 \ldots a_n \Rightarrow p/a_1$ (or) p/a_2 (or) p/a_3 (or) $\ldots$ (or) p/a_n.

② If p is not a prime, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$ is not true.

eg $ab = 16 \times 9 = 144$ 6 is not a prime

$6 \mid 144$ but $6 \nmid 9$ and $6 \nmid 16$.

Fundamental theorem of arithmetic (Unique Factorization theorem)

Every positive integer $a > 1$ can be expressed as a product of primes uniquely.

factored uniquely into a product of primes.

Every integer > 1 can be written in the form $p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$ where $n_i \geq 0$ and p_i's are distinct primes. The factorization is unique, except possibly for the order of the factors.

Proof:

Existence: We prove that a can be expressed as a product of primes.

Let $S = \{ a > 1 \mid a \in \mathbb{N}$ and a cannot be expressed as a product of primes $\}$.

If $S = \phi$ the theorem is proved.

Suppose that $S \neq \phi$.

Since $S \subset \mathbb{N}$, by well-ordering principle

S has a least member, say, m.

Every nonempty set of nonnegative integers has a smallest element

If $m = m \cdot 1$ only then m is a prime $\Rightarrow m \notin S$.

$\therefore$ m can be expressed as $m = m_1 \cdot m_2$, where $1 < m_1 < m$ and $1 < m_2 < m$.

Since m is the least member of S and $m_1, m_2 < m$

$\Rightarrow m_1 \notin S, m_2 \notin S$

$\Rightarrow m_1, m_2$ can be expressed as a product of primes

$\Rightarrow m = m_1 \cdot m_2$ can be expressed as a product of primes

This is a contradiction. $\therefore S = \phi$

Hence every integer $a > 1$ can be expressed as a product of primes.

Uniqueness: If $a = p_1 p_2 \cdots p_s = q_1 q_2 \cdots q_t$ are two prime factorizations for a, we prove that $s = t$ and every $p_i = q_j$ for some j.

We use induction on the positive integer s.

Let $s = 1$ then $p_1 = q_1 q_2 \cdots q_t$

$\Rightarrow q_1 = p_1$ and $t = 1$ ($\because p_1$ is a prime)

$\therefore$ The ~~theorem~~ statement is true for $s = 1$.

Suppose that the statement is true for $s = r$

i.e if $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_t$ then $t = r$ and every $p_i = q_j$ for some j.

Now we have to prove that the statement is true for $s = r+1$

i.e if $p_1 p_2 \cdots p_r p_{r+1} = q_1 q_2 \cdots q_t$ then $t = r+1$ and every $p_i = q_j$ for some j.

$p_1 p_2 \cdots p_r p_{r+1} = q_1 q_2 \cdots q_t \Rightarrow p_{r+1} \mid q_1 q_2 \cdots q_t$

$\Rightarrow p_{r+1} \mid q_j$ for some j.

$\Rightarrow p_{r+1} = q_j$ since p_{r+1}, q_j are primes.

$\therefore p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_{j-1} q_{j+1} \cdots q_t$

Since the statement is supposed to be true for $t = r$

$\therefore$ we have $r = t-1$ and $p_i = q_j$ for some j

i.e $t = r+1$ and $p_i = q_j$ for some j.

$\therefore$ By induction the statement is true for all positive integers s.

$\Rightarrow a = q_1 q_2 \cdots q_t$ is only a rearrangement of $a = p_1 p_2 \cdots p_s$ such that $s = t$ and $p_i = q_j$ for some j.

Note: The above representation of "a" is called prime factorization of 'a' in canonical form (or) prime power factorization of a.

↓ G.c.d. using prime factorization:

Let $\min(a,b)$ denotes the smaller (or) minimum, of the two numbers a and b.

Let the prime factorization of a and b ~~are~~ be

$a = p_1^{a_1} \cdot p_2^{a_2} \cdots p_n^{a_n}$, $b = p_1^{b_1} \cdot p_2^{b_2} \cdots p_n^{b_n}$ where each exponent is a nonnegative integer and where all primes occurring in the prime factorizations of a and b are included in both products, perhaps with zero exponents.

$$\text{G.c.d.}(a,b) = p_1^{\min(a_1,b_1)} \cdot p_2^{\min(a_2,b_2)} \cdots p_n^{\min(a_n,b_n)}$$ since

for each prime p_i, a and b share exactly $\min(a_i, b_i)$ factors of p_i.

$(15,21) = 3 \quad (24,36) = 2^2 \cdot 3 = 12$

$24 = 2^3 \cdot 3$

$36 = 2^2 \cdot 3^2$

↑

Defn. (L.C.M.)

The least common multiple of two positive integers a and b is the smallest positive integer that is divisible by a and b.

The least common multiple of a and b is denoted by $[a,b]$.

eg. $[15,21] = 105$, $[24,36] = 72$, $[2,20] = 20$ ....

$15 = 3 \times 5 \quad 21 = 3 \times 7$

$3 \times 5 \times 7 = 105$

L.C.M. of $a,b = [a,b] = p_1^{\max(a_1,b_1)} \cdot p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)}$

where $\max(x,y)$ denotes the larger (or) maximum of x and y.

Note If x and y are real numbers, then $\max(x,y) + \min(x,y) = x+y$.

Theorem: If a and b are positive integers then $[a,b](a,b) = ab$ where $[a,b]$ and (a,b) are the L.C.M. and g.c.d. of a and b resply.

Proof: Let a and b have prime power factorizations

$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$ and $b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$ where the exponents are non-negative integers and all primes occurring in either factorization occur in both, perhaps with zero exponents.

Now let $M_j = \max(a_j, b_j)$ and $m_j = \min(a_j, b_j)$. Then we have

$$[a,b](a,b) = p_1^{\max(a_1,b_1)} \cdot p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)} \cdot p_1^{\min(a_1,b_1)} \cdot p_2^{\min(a_2,b_2)} \cdots p_n^{\min(a_n,b_n)}$$

$$= p_1^{M_1+m_1} \cdot p_2^{M_2+m_2} \cdot p_3^{M_3+m_3} \cdots p_n^{M_n+m_n}$$

$$= p_1^{a_1+b_1} \cdot p_2^{a_2+b_2} \cdot p_3^{a_3+b_3} \cdots p_n^{a_n+b_n}$$

$$= (p_1^{a_1} \cdot p_2^{a_2} p_3^{a_3} \cdots p_n^{a_n})(p_1^{b_1} \cdot p_2^{b_2} \cdots p_n^{b_n})$$

$$= ab$$

$\therefore [a,b](a,b) = ab$. Since $M_j + m_j = a_j + b_j$

Lemma: Let m and n be relatively prime positive integers. Then if d is a positive divisor of mn, there is a unique pair of positive divisors d_1 of m and d_2 of n such that $d = d_1 d_2$.

Conversely, if d_1 and d_2 are positive divisors of m and n respectively, then $d = d_1 d_2$ is a positive divisor of mn.

Proof:

Let the prime-power factorizations of m and n be

$m = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s}$ and $n = q_1^{n_1} q_2^{n_2} \cdots q_t^{n_t}$.

Since $(m,n) = 1$, the set of primes $p_1, p_2, \cdots p_s$ and the set of primes $q_1, q_2, \cdots q_t$ have no common elements.

$\therefore$ The prime-power factorization of mn is

$$mn = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s} \cdot q_1^{n_1} q_2^{n_2} \cdots q_t^{n_t}$$

Hence if d is a positive divisor of mn, then, we must have

$$d = p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s} \cdot q_1^{f_1} q_2^{f_2} \cdots q_t^{f_t}$$

where $0 \le e_i \le m_i$ for $i = 1, 2, \cdots, s$ and $0 \le f_j \le n_j$ for $j = 1, 2 \cdots t$.

Assuming, $d_1 = p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s}$ and $d_2 = q_1^{f_1} q_2^{f_2} \cdots q_t^{f_t}$

d_1 and d_2 are positive divisors of m and n respectively

and $d = d_1 d_2$.

Conversely, let d_1 and d_2 be positive divisors of m and n, respectively.

Then $d_1 = p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s}$, $d_2 = q_1^{f_1} q_2^{f_2} \cdots q_t^{f_t}$

where $0 \le e_i \le m_i$ for $i = 1, 2, \cdots, s$ and $0 \le f_j \le n_j$ for $j = 1, 2, \cdots t$

$\therefore$ $d_1 d_2 = p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s} \cdot q_1^{f_1} q_2^{f_2} \cdots q_t^{f_t} = d$, say

which is clearly a positive divisor of $mn = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s} \cdot q_1^{n_1} q_2^{n_2} \cdots q_t^{n_t}$

since the power of such prime occurring in the prime power factorization of d is less than or equal to the power of that prime in the prime-power factorization of mn.

Prove that If a and b are integers both of the form $4n+1$, then product ab is also of this form.

Soln Since a and b are both of the form $4n+1$.

$\exists\ r, s \in \mathbb{Z}$ such that $a = 4r+1$, $b = 4s+1$.

Now $ab = (4r+1)(4s+1)$

$= 16rs + 4r + 4s + 1 = 4(4rs + r + s) + 1$

$= 4n_1 + 1$

$\Rightarrow$ ab is also of the form $4n+1$.

Hence proved.

Note i) $x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1})$, $n \in \mathbb{N}$.

ii) If n is odd,

$x^n + y^n = (x+y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 - \cdots + y^{n-1})$.

(n is odd $\Rightarrow$ n-1 is even)

Problem:

If n is an odd +ve integer then prove that $2^{2n}+1$ is divisible by 5.

Soln Given n is an odd +ve integer.

$\therefore 2^{2n}+1 = (2^2)^n + 1 = 4^n + 1$

$= (4+1)(4^{n-1} - 4^{n-2} + 4^{n-3} - \cdots + 1^{n-1})$

$= 5(4^{n-1} - 4^{n-2} + 4^{n-3} - \cdots + 1)$

$= 5k$.

$\Rightarrow 5 \mid 2^{2n}+1$, if n is odd

Hence proved

prove that $3^{2n+1} + 2^{n+2}$ is divisible by 7.

Soln

Let $f(n) = 3^{2n+1} + 2^{n+2}$

If $n=1$, $f(1) = 3^3 + 2^3 = 27+8 = 35$ is divisible by 7.

$\Rightarrow$ $f(n)$ is divisible by 7 if $n=1$.

ie The result (statement) is true for $n=1$

Let us assume that $f(n)$ is divisible by 7 if $n=m$.

$\Rightarrow f(m) = 7k$.

$\Rightarrow 3^{2m+1} + 2^{m+2} = 7k$

$\Rightarrow 3^{2m+1} = 7k - 2^{m+2}$ ⟶ (1)

If $n = m+1$,

$f(m+1) = 3^{2(m+1)+1} + 2^{m+1+2}$

$= 3^{2m+3} + 2^{m+3}$

$= 3^{2m+1} \cdot 3^2 + 2^{m+2} \cdot 2^1$

$= 3^2 (7k - 2^{m+2}) + 2^{m+2} \cdot 2$

$= 9(7k) - 9 \cdot 2^{m+2} + 2 \cdot 2^{m+2}$

$= 9(7k) - 7 \cdot 2^{m+2}$

$= 7(9k - 2^{m+2})$

$f(m+1) = 7k_1$, where $k_1 = 9k - 2^{m+2}$

$\Rightarrow f(m+1)$ is divisible by 7

$\Rightarrow$ The result is true for $n = m+1$. Hence proved by induction

Note:: The product of r consecutive integers is divisible by $r!$

For $x \in \mathbb{N}$, $P = (x+1)(x+2)\cdots(x+r) = \frac{(x+r)!}{x!} \Rightarrow \frac{P}{r!} = \frac{(x+r)!}{x!\,r!} = {}^{(x+r)}C_r = +ve$ integer

show that

i) The sum of an integer and its square is an even

ii) The difference b/w the square of an integer and itself is an even

iii) If m and n are +ve integers then $(m+n)!$ is divisible by $m!\,n!$

iv) If n is an integer then (a) $n(n^2-1)$ is a multiple of 6

(b) $n(n+1)(n+5)$ " " "

(c) $n(n+1)(2n+1)$ " " "

(d) $n(n-1)(2n-1)$ " " "

Dirichlet's Theorem on primes in Arithmetic Progressions:

Let a and b be relatively prime positive integers. Then the arithmetic progression $an+b$, $n = 1, 2, 3, \ldots$ contains infinitely many primes.

Proposition: There are infinitely many primes of the form $4n+3$, where n is a positive integer.

property: If a, m, n are non negative integers then $(a, mn) = 1 \iff$ $(a, m) = 1$ & $(a, n) = 1$

Sol'n $\Leftarrow$: Let $(a, m) = 1$ & $(a, n) = 1$

$\because (a, m) = 1 \Rightarrow \exists\ h, k \in \mathbb{Z} \ni ah + mk = 1 \longrightarrow ①$

$\because (a, n) = 1 \Rightarrow \exists\ r, t \in \mathbb{Z} \ni ar + nt = 1 \Rightarrow$

$ar + nt \cdot 1 = 1 \longrightarrow ②$

putting the value of 1 from eqn ① in L.H.S of ②,

we have $ar + nt(ah + mk) = 1$

$\Rightarrow ar + ahnt + mntk = 1$

$\Rightarrow a(r + hnt) + (mn)tk = 1$

$\Rightarrow (a, mn) = 1$

$\Rightarrow$: Let $(a, mn) = 1$

$\therefore \exists\ u, v \in \mathbb{Z} \ni au + mnv = 1$

$\Rightarrow a(u) + m(nv) = 1$

$\therefore (a, m) = 1$

Illy $(a, n) = 1$

If a and b are any two odd primes, show that $a^2 - b^2$ is a Composite number.

Sol'n $a^2 - b^2 = (a - b)(a + b)$

$\because$ a and b are odd primes

So let $a = 2k + 1$, $b = 2k' + 1$

$\therefore a-b = 2k+1-(2k'+1) = 2(k-k')$ is even

$a+b = 2k+1+2k'+1 = 2(k+k'+1)$ is even

$\therefore$ Neither $a-b$ nor $a+b$ is 1

$\therefore a^2-b^2$ is a composite number.

If a^2-b^2 is a prime number, show that $a^2-b^2 = a+b$ where a, b are natural numbers.

Soln Given a^2-b^2 is a prime number

Now $a^2-b^2 = (a+b)(a-b)$

$\Rightarrow a-b = 1$ ($\because a+b > 1$, $a, b \in \mathbb{N}$ then $a-b < a+b \Rightarrow a+b > 1$)

$\therefore a^2-b^2 = (a+b)1 = a+b$.

Problem: (a) If r is an integer, S.T. $r(r^2-1)(3r+2)$ is divisible by 24

(b) If n is even, P.T. $n(n+1)(n+2)$ is divisible by 24

(c) If n is odd, S.T. $n(n^2-1)$ is divisible by 24

Proof: (a) $r(r^2-1)(3r+2) = r(r-1)(r+1)(3(r+2)-4)$

$= 3(r-1)r(r+1)(r+2) - 4(r-1)r(r+1)$

($\because$ product of r consecutive integers is divisible by $r!$)

$= 3.4!k_1 - 4(3!)k_2$ where $k_1, k_2 \in \mathbb{Z}$

$= 72k_1 - 24k_2$

$= 24(3k_1 - k_2)$

$= 24k$ where $k = 3k_1 - k_2 \in \mathbb{Z}$

$\Rightarrow r(r^2-1)(3r+2)$ is divisible by 24

(b) Given n is even let $n = 2m$, $m \in \mathbb{Z}$

$n(n+1)(n+2) = 2m(2m+1)(2m+2) = 2m(2m+1)2(m+1)$

$= 4m(m+1)(2m+1)$

$= 4.6k$

$= 24K \quad K \in \mathbb{Z}$

(c) If (b) Given n is odd let $n = 2m+1$, $m \in \mathbb{Z}$

$$n(n^2-1) = n(n+1)(n-1)$$

$$= (2m+1)(2m+1+1)(2m+1-1)$$

$$= (2m+1)(2m+2)\,2m$$

$$= 2m \;\; 2(m+1)(2m+1)$$

$$= 4m(m+1)(2m+1) = 4 \cdot 6k = 24k, \; k \in \mathbb{Z}$$

$\therefore n(n^2-1)$ is divisible by 24 if n is odd

Prove that 4 does not divide m^2+2 for any integer m.

soln $\because$ m is an integer

$\therefore$ Either m is even (or) m is odd

Case ① If m is even let $m = 2k$.

$$\therefore m^2+2 = (2k)^2+2 = 4k^2+2 = 2(2k^2+1)$$

$$= 2 \times \text{odd integer}$$

which is not divisible by 4

Case ② If m is odd, let $m = 2k+1$

$$\therefore m^2+2 = (2k+1)^2+2 = 4k^2+4k+1+2$$

$$= 2[(2k^2+2k)+1]+1$$

$$= 2 \times \text{odd integer}$$

$\therefore$ It is not divisible by 4

$\therefore$ 4 does not divide m^2+2 for any $m \in \mathbb{Z}$

If $n > 2$, show that $n^5 - 5n^3 + 4n$ is divisible by 120

(a) $n^5 - 5n^3 + 4n = n(n^4 - 5n^2 + 4)$

$$= n(n^4 - 4n^2 - n^2 + 4)$$

$$= n(n^2(n^2-1) - 4(n^2-1))$$

$\Rightarrow n^5 - 5n^3 + 4n = n(n^2-1)(n^2-4)$

$= (n-2)(n-1)n(n+1)(n+2)$

$= 5!k = 120k \quad k \in \mathbb{Z}$

$\therefore n^5 - 5n^3 + 4n$ is divisible by 120 if $n > 2$

If $n=1$, $n^5 - 5n^3 + 4n = 1 - 5 + 4 = 0$

If $n=2$, $32 - 40 + 8 = 0$

Prove that $9^n - 8^n - 1$ is divisible by 8

Soln Let $f(n) = 9^n - 8^n - 1$

If $n=1$, $f(1) = 9 - 8 - 1 = 0$ is divisible by 8

$\therefore f(n)$ is ~~true~~ if $n=1$

Let us assume that $f(n)$ is divisible by 8 if $n = m$.

$\Rightarrow f(m) = 8k$

$\Rightarrow 9^m - 8^m - 1 = 8k$

$\Rightarrow 9^m = 8^m + 1 + 8k \longrightarrow (1)$

If $n = m+1$ then $f(m+1) = 9^{m+1} - 8^{m+1} - 1$

$= 9 \cdot 9^m - 8^m \cdot 8 - 1$

$= 9(8^m + 1 + 8k) - 8 \cdot 8^m - 1$

$= 8^m + 8 + 9(8k)$

$= 8(9k + 1 + 8^{m-1})$

$= 8k_1$, where $k_1 = 9k + 1 + 8^{m-1}$

$\longrightarrow f(m+1)$ is divisible by 8.

$\Rightarrow$ The statement is true for $n = m+1$

∴ By induction, $f(n)$ is divisible by 3 for all $n \in \mathbb{N}$.

oblem:

If 2^n+1 is odd prime, then prove that n is a power of 2.

ɔlⁿ. Every natural number n can be expressed as $n = 2^k q$

where q is odd, $k \geq 0$.

$\mathbb{N} = \{1, 2, 3, \ldots\}$

$n \in \mathbb{N}$

$n = 1 = 2^0 \cdot 1$

$n = 2 = 2^1 \cdot 1$

$n = 3 = 2^0 \cdot 3$

$n = 4 = 2^2 \cdot 1$

$n = 5 = 2^0 (5)$

If possible assume that $q \neq 1$.

Since $q \neq 1$ and q is odd $\Rightarrow q > 1$

we have $n = 2^k q$

$\Rightarrow n = mq$ where $m = 2^k$, $q > 1$

$m = 2^k \geq 1$

Now $2^n + 1 = 2^{mq} + 1$

$= (2^m)^q + 1^q$

$= (2^m + 1)\left((2^m)^{q-1} - (2^m)^{q-2} + \ldots + 1^{q-1}\right)$

$\Rightarrow 2^n + 1 = (2^m + 1)\left((2^m)^{q-1} - (2^m)^{q-2} + \ldots + 1^{q-1}\right)$

∴ $2^n + 1$ is a composite number as $2^m + 1 \geq 2$

which is a contradiction to $2^n + 1$ is odd prime

$\Rightarrow q \neq 1$ is false

$\Rightarrow q = 1$ is true

∴ $n = 2^k$ where $k \geq 0$.

∴ n is a power of 2. Hence proved

P.T. an integer is divisible by 9 if and only if sum of its digits is divisible by 9. (15)

Soln Let a be any integer. ie ~~let a be an n digit number~~

$\Rightarrow a = a_n a_{n-1} a_{n-2} \cdots a_1 a_0$

$a = a_n(10^n) + a_{n-1}(10^{n-1}) + a_{n-2}10^{n-2} + \cdots + a_1(10) + a_0$

$\Rightarrow a = a_0 + a_1(10) + 100a_2 + 1000a_3 + \cdots + 10^n a_n$

$= a_0 + (a_1 + 9a_1) + (a_2 + 99a_2) + (a_3 + 999a_3) + \cdots + 10^n a_n$

$= (a_0 + a_1 + a_2 + \cdots + a_n) + 9(a_1 + 11a_2 + 111a_3 + \cdots + \cdots)$

$a = S + 9k$ where S = sum $= a_0 + a_1 + a_2 + \cdots + a_n$

$k = a_1 + 11a_2 + 111a_3 + \cdots + \cdots$

$\Rightarrow a - S = 9k$

$\Rightarrow 9 \mid a - S \longrightarrow (1)$

Case (1)

If an integer 'a' is divisible by 9 then To prove that S is divisible by 9 we have $9 \mid a$ and $9 \mid a - s$ from (1)

$\Rightarrow 9 \mid a - (a - s)$

$\Rightarrow 9 \mid s$

$\Rightarrow$ Sum of the digits is divisible by 9

Case (2)

If sum of the digits is divisible by 9

Then to prove that a is divisible by 9

we have $9 \mid s$ and $9 \mid a-s$ from (1)

$$\Rightarrow 9 \mid a-s+s$$

$$\Rightarrow 9 \mid a$$

$\therefore$ a is divisible by 9

Hence proved

Prove that the number of primes is infinite.

Soln Suppose that the number of primes is finite.

Let $S = \{p_1, p_2, \ldots p_n\}$ be the set of all primes.

Take the positive integer $N = p_1 p_2 \ldots p_n + 1$.

Since $N > 1$, N has a prime factor

For each $p_i \in S$, we can write $N = M p_i + 1$

where $M = p_1 p_2 \ldots p_{i-1} p_{i+1} \ldots p_n$

$\therefore$ when N is divided by each $p_i \in S$, we have remainder $= 1$.

Thus each $p_i \in S$ is not a factor of N

$\therefore N > 1$ has no prime factor as the set S contains all the primes

This is a Contradiction.

Hence, the number of primes is infinite

If $p = 2^n - 1$ is a prime then prove that n is a prime.

Soln We have $p = 2^n - 1$ is a prime

Suppose that n is not a prime.

$\therefore \exists\ r, s \in \mathbb{Z}$ such that $n = rs$ where $1 < r, s < n$.

Now $p = 2^n - 1 = 2^{rs} - 1 = (2^r)^s - 1^s$

$= (2^r - 1)\left((2^r)^{s-1} + (2^r)^{s-2} + \cdots + 1^{s-1}\right)$

$\Rightarrow p = (2^r - 1)\left((2^r)^{s-1} + (2^r)^{s-2} + \cdots + 1\right)$

$\Rightarrow p$ is a composite number. ($\because 2^r > 2$ as $r > 1$ $\Rightarrow 2^r - 1 > 1$)

which is a contradiction to $p = 2^n - 1$ is a prime

$\therefore n$ is a prime.

The number of divisors of a positive integer N and the sum of all the distinct positive integral divisors of a positive integer.

By Fundamental theorem of arithmetic,

A positive integer $N > 1$ can be written as a product of primes uniquely

i.e. $\therefore N = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ where $1 < p_1 < p_2 < p_3 < \cdots < p_r$ and

$\alpha_1, \alpha_2, \cdots \alpha_r$ are positive integers.

Now, Consider the product,

$$P = (1 + p_1 + p_1^2 + \cdots + p_1^{\alpha_1})(1 + p_2 + p_2^2 + \cdots + p_2^{\alpha_2}) \cdots (1 + p_r + p_r^2 + \cdots + p_r^{\alpha_r})$$

General term of this product is $p_1^{\beta_1} \cdot p_2^{\beta_2} \cdots p_r^{\beta_r}$ where

$$0 \le \beta_1 \le \alpha_1,\ 0 \le \beta_2 \le \alpha_2,\ \cdots,\ 0 \le \beta_r \le \alpha_r$$

$\therefore p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ is a factor (or) divisor of $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r} = N$.

Con.

Hence, we can say that every factor (divisor) of N is a term of P.

Hence (i) The number of factors (divisors) of N = The number of terms of P

$$= (1+\alpha_1)(1+\alpha_2)\cdots(1+\alpha_r)$$

$\therefore$ The number of factors (divisors) of N $= (1+\alpha_1)(1+\alpha_2)\cdots(1+\alpha_r)$ and is denoted by $\tau(N)$

ii) The sum of all the distinct divisors of N

= The sum of all the terms of P

$$= (1+p_1+p_1^2+p_1^3+\cdots+p_1^{\alpha_1})(1+p_2+p_2^2+\cdots+p_2^{\alpha_2})\cdots(1+p_r+p_r^2+\cdots+p_r^{\alpha_r})$$

$$= \left(\frac{p_1^{\alpha_1+1}-1}{p_1-1}\right)\left(\frac{p_2^{\alpha_2+1}-1}{p_2-1}\right)\cdots\left(\frac{p_r^{\alpha_r+1}-1}{p_r-1}\right)$$

The sum of all the distinct positive integral divisors of $N>1$ is denoted by $\sigma(N)$.

Find the number of divisors and sum of the divisors of 14553.

Soln $14553 = 3^3\times 539 = 3^3\times 7^2\times 11^1$

Here $p_1=3,\ p_2=7,\ p_3=11$

$\alpha_1=3,\ \alpha_2=2,\ \alpha_3=1$

Num of divisors of $14553 = \tau(14553) = (1+\alpha_1)(1+\alpha_2)(1+\alpha_3) = 4(3)(2)$

$= 24$.

Sum of divisors of $14553 = \sigma(14553) = \left(\frac{p_1^{\alpha_1+1}-1}{p_1-1}\right)\left(\frac{p_2^{\alpha_2+1}-1}{p_2-1}\right)\left(\frac{p_3^{\alpha_3+1}-1}{p_3-1}\right)$

$$= \left(\frac{3^4-1}{3-1}\right)\left(\frac{7^3-1}{7-1}\right)\left(\frac{11^2-1}{11-1}\right)$$

$= 40\times 57\times 12 = 27360$.

Perfect number.

If the sum of all divisors of $n > 1$, is equal to $2n$, then n is called a perfect number. (17)

eg $n = 28 = 7 \times 2^2$ is perfect number

since, $\sigma(28) = \left(\frac{7^2-1}{7-1}\right)\left(\frac{2^3-1}{2-1}\right) = \frac{48}{6}\left(\frac{7}{1}\right) = 56 = 2(28)$

Bracket function (or) Step function (or) Integral part function.

Defn. The function $I : \mathbb{R} \to \mathbb{Z}$ defined by $I(x) = n$ where $n \le x < n+1$, is called the bracket function (or) the integral part function.

Integral part of $x \in \mathbb{R}$ is denoted by $I(x)$ (or) $[x]$.

Defn. If $x \in \mathbb{R}$, $x - [x]$ is called the Fractional part of x.

Note (1) $[x] \le x < [x]+1$ (or) $x - 1 < [x] \le x$.

(2) For every $x \in \mathbb{R}$, $x \ge [x] \Rightarrow x - [x] \ge 0$

i.e The Fractional part of any $x \in \mathbb{R}$ is non negative.

(3) $x \in \mathbb{Z} \Leftrightarrow [x] = x$

(4) $x, y \in \mathbb{R}$ then $[x+y] \ge [x] + [y]$

(5) If $x \in \mathbb{R}$ then $[x] + [-x] = \begin{cases} 0 & \text{if } x \in \mathbb{Z} \\ -1 & \text{if } x \notin \mathbb{Z} \end{cases}$.

(6) If $a \in \mathbb{Z}$, $b \in \mathbb{N}$ then $\left[\frac{a}{b}\right]$ = the quotient when a is divided by b

(7) For $x \in \mathbb{R}$, $[x] \le x < [x]+1 \Rightarrow x$ lies b/w $[x]+0$ and $[x]+1$

$\Rightarrow x = [x] + \theta$ where $0 \le \theta < 1$.

(8) If p is a prime number, then the highest power of p contained in $n!$ is $\sum_{r=1}^{\infty} \left[\frac{n}{p^r}\right]$.

problem: (1) Find the highest power of 3 which is contained in 100!

soln $\sum_{r=1}^{\infty} \left[\frac{100}{3^r}\right] = \left[\frac{100}{3}\right] + \left[\frac{100}{3^2}\right] + \left[\frac{100}{3^3}\right] + \left[\frac{100}{3^4}\right] + \left[\frac{100}{3^5}\right] + \cdots$

$= 33 + 11 + 3 + 1 + 0 + 0 + \cdots$

$= 48$

The highest power of 3 contained in 100! is 48

i.e. $100! = 3^{48} \cdot k \quad , k \in \mathbb{Z}$.

(2) Find the highest power of 5 that will divide $27.28.29 \cdots 100$

soln

$27.28.29 \cdots 100 = \frac{100!}{26!}$

The highest power of 5 contained in $100! = \sum_{r=1}^{\infty} \left[\frac{100}{5^r}\right]$

$= \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] + \left[\frac{100}{5^3}\right] + \cdots$

$= 20 + 4 + 0$

$= 24$

$\therefore 100! = 5^{24} k_1$ where $k_1 \in \mathbb{Z}$

The highest power of 5 contained in $26! = \sum_{r=1}^{\infty} \left[\frac{26}{5^r}\right] = \left[\frac{26}{5}\right] + \left[\frac{26}{5^2}\right] + \left[\frac{26}{5^3}\right] + \cdots$

$= 5 + 1 + 0 = 6$

$\therefore 26! = 5^6 k_2$ where $k_2 \in \mathbb{Z}$

$\frac{100!}{26!} = \frac{5^{24} k_1}{5^6 k_2} = \frac{5^{18} k_1}{k_2}$

∴ The highest power of 5 that will divide $27.28.29...100 =$

$= 24 - 6 = 18.$ (18)

Ex ① Find the prime factorization of 20!

② How many zeros are there at the end of 1000! in decimal notation?

Mersenne number:

A number of the form $M_n = 2^n - 1$ where n is a prime is called Mersenne number.

Note: All the Mersenne numbers need not be prime numbers.

eg $n = 11$ is a prime $M_{11} = 2^{11} - 1 = 2048 - 1 = 2047$ is divisible by 23

Fermat number:

For every non negative integer k, $F_k = 2^{2^k} + 1$ is called k^{th}-Fermat number.

Property: ① Every prime divisor of the Fermat number $F_n = 2^{2^n} + 1$ is of the form $2^{n+2}.k + 1$

② Let m and n be distinct nonnegative integers then $(F_m, F_n) = 1$.

③ Let $F_k = 2^{2^k} + 1$ denote the k^{th} Fermat number, where k is a nonnegative integer then for all positive integers n, we have

$$F_0 F_1 F_2 \cdots F_{n-1} = F_n - 2$$

68, 73, 62, 50

Equivalence Relation:

A relation R defined on a set A is called an Equivalence relation if R satisfies the following

- Reflexive : $(a,a) \in R \quad \forall\ a \in A$
- Symmetric : $(a,b) \in R \Rightarrow (b,a) \in R \quad \forall\ a,b \in A$
- Transitive : $(a,b) \in R$, and $(b,c) \in R \Rightarrow (a,c) \in R \quad \forall\ a,b,c \in A$

Congruences

The property of Congruence provides a way of classifying integers according to the remainder obtained upon division by a fixed positive integer.

Defⁿ: Congruence:

Let m be a fixed positive integer and $a, b \in \mathbb{Z}$. 'a' is said to be Congruent to 'b' modulo m, if $m \mid a-b$

If a is congruent to b modulo m, we write $a \equiv b \pmod{m}$.

If $m \nmid a-b$ then we say that a and b are incongruent modulo m and we write it as $a \not\equiv b \pmod{m}$

Note ① $a \equiv b \pmod{m} \iff m \mid (a-b) \iff a-b = qm$, $q \in \mathbb{Z}$

$\iff a-b$ is a multiple of m i.e. $a-b = M(m)$

eg: $22 \equiv 4 \pmod{9}$ i.e. $a-$ since $9 \mid (22-4) = 18$

$33 \equiv 1 \pmod{2}$ $\quad 200 \equiv 2 \pmod{9}$

Note ② The congruence relation has properties similar to equality relation.

Theorem: Two integers a and b are congruent modulo m iff they leave the same remainder when divided by m.

Proof let $a \equiv b \pmod{m}$

To prove that a, b have the same remainder upon division by m.

let $a = mq_1 + r_1$ where $0 \le r_1 < m$ and $b = mq_2 + r_2$ where $0 \le r_2 < m$

$\therefore a - b = m(q_1 - q_2) + (r_1 - r_2)$

$a \equiv b \pmod m \Rightarrow m \mid a-b$

$\Rightarrow m \mid m(q_1 - q_2) + (r_1 - r_2)$

$\Rightarrow m \mid (r_1 - r_2)$

$\Rightarrow r_1 - r_2 = 0$ since $r_1 - r_2 < m$

$\Rightarrow r_1 = r_2$

Let a, b have the same remainder upon division by m.

We prove that $a \equiv b \pmod m$

Let $a = mq_1 + r$, $b = mq_2 + r$ where $0 \le r \le m$.

$\therefore a - b = m(q_1 - q_2) = mq$ where $q = q_1 - q_2 \in \mathbb{Z}$

$\therefore m \mid a-b$

$\therefore a \equiv b \pmod m$

Theorem: For a fixed integer $m > 0$ the relation $a \equiv b$ of congruence modulo m i.e $a \equiv b \pmod m$ is an equivalence relation on the set of integers $\mathbb{Z}$

Proof:

We show that the set, $S = \{(a,b) \mid a \equiv b \pmod m\} \subset \mathbb{Z} \times \mathbb{Z}$ defines a relation that is reflexive, symmetric and transitive.

1. Let $a \in \mathbb{Z}$

$m \mid 0 \Rightarrow m \mid a-a \Rightarrow a \equiv a \pmod m$

$\therefore (a,a) \in S \quad \forall a \in \mathbb{Z}$

The relation is a reflexive relation.

let $(a,b) \in S \Rightarrow a \equiv b \pmod m$

$\Rightarrow m \mid (a-b) \Rightarrow m \mid -(a-b) \Rightarrow m \mid b-a$

$\Rightarrow b \equiv a \pmod m$

$\Rightarrow (b,a) \in S$ ∴ The relⁿ is a symmetric relⁿ

3. let $a,b,c \in \mathbb{Z} \ni (a,b) \in S, (b,c) \in S$

$(a,b) \in S \Rightarrow a \equiv b \pmod m \Rightarrow m \mid (a-b)$

$(b,c) \in S \Rightarrow b \equiv c \pmod m \Rightarrow m \mid b-c$

$\Rightarrow m \mid a-b+b \mp c$

$\Rightarrow m \mid a-c$

$\Rightarrow a \equiv c \pmod m$

$\Rightarrow (a,c) \in S$. ∴ The relation is a transitive relation.

Hence "Congruence modulo m" relation is an equivalence relation on $\mathbb{Z}$.

Theorem: If a,b,c and m are integers with $m>0$ such that $a \equiv b \pmod m$ then (i) $a+c \equiv (b+c) \pmod m$ (ii) $a-c \equiv (b-c) \pmod m$ (iii) $ac \equiv bc \pmod m$

Proof (i) $a \equiv b \pmod m \Rightarrow a-b = q_1 m,\ q_1 \in \mathbb{Z}$ (or) $m \mid a-b$

$\Rightarrow m \mid (a+c)-(b+c)$ for $c \in \mathbb{Z}$

$\Rightarrow a+c \equiv (b+c) \pmod m$

ii) $a \equiv b \pmod m \Rightarrow m \mid a-b \Rightarrow m \mid (a-c)-(b-c) \Rightarrow a-c \equiv (b-c) \pmod m$ $\forall c \in \mathbb{Z}$

iii) $a \equiv b \pmod m \Rightarrow m \mid a-b \Rightarrow m \mid c(a-b)\ \forall c \in \mathbb{Z}$

$\Rightarrow m \mid ac-bc \Rightarrow ac \equiv bc \pmod m$

Note. If $ac \equiv bc \pmod m$ then it does not imply that $a \equiv b \pmod m$

eg $14 = 7 \cdot 2 \equiv 8 \pmod 6$ i.e. $14 \equiv 8 \pmod 6$

$\Rightarrow 2 \cdot 7 \equiv 2 \cdot 4 \pmod 6$

but $7 \not\equiv 4 \pmod 6$.

Theorem: If a, b, c and m are integers with $m > 0$ and $ac \equiv bc \pmod m$ and $(c, m) = d$ then $a \equiv b \pmod{\frac{m}{d}} = b \pmod{\frac{m}{(c,m)}}$

Proof We have $ac \equiv bc \pmod m$

$\Rightarrow m \mid ac - bc \Rightarrow m \mid c(a-b)$

$\Rightarrow c(a-b) = km$, where $k \in \mathbb{Z}$

$\Rightarrow \frac{c}{d}(a-b) = k\left(\frac{m}{d}\right) \Rightarrow (m/d) \mid \frac{c}{d}(a-b)$

As $(c, m) = d \Rightarrow \left(\frac{c}{d}, \frac{m}{d}\right) = 1$

From Gauss lemma, $\left(\frac{m}{d}\right) \mid a-b$

$\Rightarrow a \equiv b \left(\mod \left(\frac{m}{d}\right)\right)$

eg. Since $50 \equiv 20 \pmod{30}$

$\Rightarrow 5 \times 10 \equiv 5 \times 4 \pmod{30}$

$\Rightarrow 10 \equiv 4 \left(\mod \left(\frac{30}{5}\right)\right) \equiv 4 \pmod 6$

Here $(5, 30) = 5$

Note If $ac \equiv bc \pmod m$ where $a, b, c, m \in \mathbb{Z}$ and $m > 0$ and $(c, m) = 1$ then $a \equiv b \pmod m$

Theorem: If $a, b, c, m \in \mathbb{Z}$ with $m > 0$ such that $a \equiv b \pmod m$ and $c \equiv d \pmod m$ then i) $a + c \equiv (b + d) \pmod m$

ii) $a - c \equiv (b - d) \pmod m$

iii) $ac \equiv bd \pmod m$.

Proof: we have $a, b, c, d \in \mathbb{Z}$, $m \in \mathbb{Z}$ with $m > 0$

$a \equiv b \pmod m \Rightarrow m \mid a-b \Rightarrow a-b = mq_1$, where $q_1 \in \mathbb{Z}$

$c \equiv d \pmod m \Rightarrow m \mid c-d \Rightarrow c-d = mq_2$ where $q_2 \in \mathbb{Z}$

ii)

$\Rightarrow (a-b) - (c-d) = m(q_1 - q_2)$

$\Rightarrow (a-c) - (b-d) = m(q_1 - q_2) = mq$

where $q = q_1 - q_2 \in \mathbb{Z}$

$\Rightarrow m \mid (a-c) - (b-d)$

$\Rightarrow a-c \equiv (b-d) \pmod m$

i) $a-b+(c-d) = m(q_1+q_2) \Rightarrow (a+c) - (b+d) = m(q_1+q_2)$

$= mk$

where $k = q_1 + q_2 \in \mathbb{Z}$

$\Rightarrow m \mid (a+c) - (b+d)$

$\Rightarrow a+c \equiv (b+d) \pmod m$

iii) $ac - bd = a(mq_1 + b)(d + mq_2) - bd = mq_1 d + m^2 q_1 q_2 + bd + bmq_2 - bd$

$\Rightarrow ac - bd = mq_1 d + m^2 q_1 q_2 + bmq_2 = m(q_1 d + q_2 b + mq_1 q_2)$

$= mq'$ where

$q' = q_1 d + q_2 b + mq_1 q_2$

$\therefore ac \equiv bd \pmod m$

Theorem: If a, b, k and m are integers with $k \geq 1$, $m > 0$ and if $a \equiv b \pmod m$ then $a^k \equiv b^k \pmod m$ but $a^k \equiv b^k \pmod m$ $\forall k \geq 2$ then $a \equiv b \pmod m$ may not be true.

Soln. $a \equiv b \pmod m \Rightarrow m \mid (a-b)$

For $k \geq 1$, $a^k - b^k = (a-b)(a^{k-1} + a^{k-2}b + \cdots + b^{k-1})$

$\Rightarrow (a-b) \mid (a^k - b^k) \Rightarrow m \mid a^k - b^k \Rightarrow a^k \equiv b^k \pmod m$

(By transitive property)

Note: In general, if $a_1 \equiv b_1 \pmod{m}$, $a_2 \equiv b_2 \pmod{m}$, $\cdots$ $a_n \equiv b_n \pmod{m}$

then i) $\sum_{r=1}^{n} a_r \equiv \sum_{r=1}^{n} b_r \pmod{m}$

ii) $\prod_{r=1}^{n} a_r \equiv \prod_{r=1}^{n} b_r \pmod{m}$

Theorem: If $a \equiv b \pmod{m_1}$, $a \equiv b \pmod{m_2}$, $\cdots$ $a \equiv b \pmod{m_k}$ where $a, b, m_1, m_2, \cdots m_k$ are integers with $m_1, m_2, \cdots m_k$ are positive then

$$a \equiv b \pmod{[m_1, m_2, m_3, \cdots, m_k]}$$

where $[m_1, m_2, \cdots, m_k]$ is the L.C.M. of $m_1, m_2, \cdots m_k$.

Proof: Since $a \equiv b \pmod{m_1}$, $a \equiv b \pmod{m_2}$, $\cdots$ $a \equiv b \pmod{m_k}$

We know that $m_1 \mid a-b$, $m_2 \mid a-b$, $\cdots$ $m_k \mid a-b$

$\therefore [m_1, m_2, \cdots m_k] \mid a-b$

$\Rightarrow a \equiv b \pmod{[m_1, m_2, \cdots m_k]}$

Note: If $a \equiv b \pmod{m_1}$, $a \equiv b \pmod{m_2}$ $\cdots$ $a \equiv b \pmod{m_k}$ where a, b are integers and $m_1, m_2, \cdots m_k$ are relative prime positive integers then $a \equiv b \pmod{m_1 m_2 \cdots m_k}$ (LCM × GCD = product)

Equivalence class:

If A is a set and if "$\sim$" (or) R is an equivalence ~~relation~~ relation on A, then the equivalence class of $a \in A$ is the set $\{x \in A \mid a \sim x\}$.

We write it as $cl(a)$

i.e $cl(a) = \{x \in A \mid a \sim x\}$

Complete residue system:

Let m be a fixed positive integer. Then a set $\{a_1, a_2, \cdots a_k\}$ of integers is called a complete residue system modulo m if

i) $a_i \not\equiv a_j \pmod m$ for each $i \neq j$ and

ii) for each integer n There exist unique a_i such that

$n \equiv a_i \pmod m$.

(OR)

The remainder r, upon division of x by m, is called the residue of $x \bmod m$. The set of integers $\mathbb{Z}_m = \{0, 1, 2, 3, \cdots, (m-1)\}$ called the set of the complete system of residue modulo m.

(The set of the least positive residues modulo m).

Complete set of residue classes:

The complete set of residue classes modulo m is the collection $\{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \cdots, (\overline{m-1})\}$ where $\bar{r} = \{x / x \equiv r \pmod m\}$

$$= \{x / x - r = mk, k \in \mathbb{Z}\}$$

$$= \{x / x = r + mk, k \in \mathbb{Z}\}$$

$$\bar{r} = \{r + mk / k \in \mathbb{Z}\}$$

$$\Rightarrow \bar{r} = \{\cdots\cdots, r-2m, r-m, r, r+m, r+2m, r+3m, \cdots\cdots\cdots\}$$

$$= \{\cdots\cdots, -2m+r, -m+r, r, m+r, 2m+r, \cdots\cdots\}.$$

An equivalence relation on a set splits the set into a number of subsets. Since congruence modulo m is an equivalence relation on $\mathbb{Z}$, this relation partitions $\mathbb{Z}$ into a collection of disjoint subsets "called residue classes" (or) "Congruence classes".

If m is a positive integer, then there exist exactly m equivalence classes for the equivalence relation "Congruence modulo m". The equivalence class $\bar{r}$ (or) $\{r\}$ (or) $cl(r)$ is the set $\{x \in \mathbb{Z} \mid x \equiv r \pmod m\}$. It is also called r- residue class (or) r- Congruence class (or) $cl(r)$. The set of m equivalence classes (or) residue classes (or) Congruence classes is denoted by

$$\mathbb{Z}_m = \{\bar{0}, \bar{1}, \bar{2}, \ldots (\overline{m-1})\}$$

For $m=6$, the complete set of residue classes modulo m is the set $\mathbb{Z}_6 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$. where $\bar{r} = \{r+6k \mid k \in \mathbb{Z}\}$

$\bar{0} = \{0+6k \mid k \in \mathbb{Z}\} = \{\ldots, -12, -6, 0, 6, 12, 18, \ldots\}$

$\bar{1} = \{1+6k \mid k \in \mathbb{Z}\} = \{\ldots -17, -11, -5, 1, 7, 13, 19, \ldots\}$

$\bar{2} = \{2+6k \mid k \in \mathbb{Z}\} = \{\ldots, -16, -10, -4, 2, 8, 14, 20, \ldots\}$

$\bar{3} = \{3+6k \mid k \in \mathbb{Z}\} = \{\ldots, -15, -9, -3, 3, 9, 15, 21, \ldots\}$

$\bar{4} = \{4+6k \mid k \in \mathbb{Z}\} = \{\ldots, -14, -8, -2, 4, 10, 16, 22, \ldots\}$

$\bar{5} = \{5+6k \mid k \in \mathbb{Z}\} = \{\ldots, -13, -7, -1, 5, 11, 17, 23, \ldots\}$

$\bar{6} = \{6+6k \mid k \in \mathbb{Z}\} = \{\ldots, -12, -6, 0, 6, 12, 18, \ldots\}$

Here $\bar{0} = \bar{6} = \bar{12} = \bar{18} = \ldots = \overline{6k}$, $k \in \mathbb{Z}$

$\bar{1} = \bar{7} = \bar{13} = \bar{19} = \cdots = \overline{6k+1}$

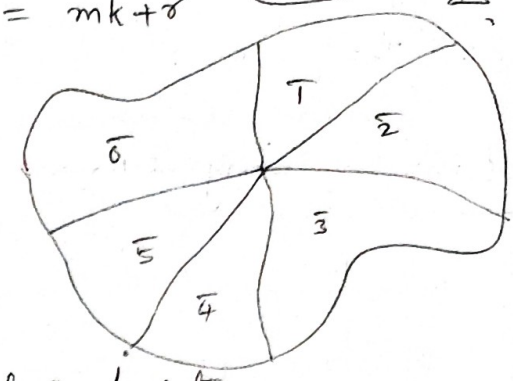
$\bar{2} = \bar{8} = \bar{14} = \bar{20} = \cdots = \overline{6k+2}$

$\bar{3} = \bar{9} = \bar{15} = \bar{21} = \cdots = \overline{6k+3}$

$\bar{4} = \bar{10} = \bar{16} = \bar{22} = \cdots = \overline{6k+4}$

$\bar{5} = \bar{11} = \bar{17} = \bar{23} = \cdots = \overline{6k+5}$

i.e $\overline{m} = \overline{mk+r}$ $(m=6)$



and $\bar{0} \cup \bar{1} \cup \bar{2} \cup \bar{3} \cup \bar{4} \cup \bar{5} = \mathbb{Z}$ and any two distinct residue classes are disjoint sets.

Find the remainder in division of 41^{65} by 7. (The least positive residue of $41^{65} \bmod 7$)

Soln Since $7 \mid 42 \Rightarrow 7 \mid 41-(-1) \Rightarrow 41 \equiv -1 \pmod 7$

$\Rightarrow 41^{65} \equiv (-1)^{65} \pmod 7$

$\Rightarrow 41^{65} \equiv -1 \pmod 7$

$\Rightarrow 41^{65} \equiv 6 \pmod 7$

$\bar{6} = 6+7k$

$6-7$

-1

∴ The remainder is 6.

Find the remainder in the division of 2^{20} is devided by 7

Soln: $7 \mid 2^3 - 1 \Rightarrow 2^3 \equiv 1 \pmod 7$

$\Rightarrow (2^3)^6 \equiv 1^6 \pmod 7$

$\Rightarrow 2^{18} \equiv 1 \pmod 7$

$\Rightarrow 2^{18} \cdot 2^2 \equiv 1 \cdot 2^2 \pmod 7$

$\Rightarrow 2^{20} \equiv 4 \pmod 7$

∴ The remainder is 4

Find the remainder when 3^{40} is divided by 23.

Soln $3^3 \equiv 4 \pmod{23}$

$\Rightarrow (3^3)^3 \equiv 4^3 \pmod{23}$

$\Rightarrow 3^9 \equiv 64 \pmod{23} \equiv -5 \pmod{23}$

$\Rightarrow (3^9)^2 \equiv (-5)^2 \pmod{23} \equiv 2 \pmod{23}$

$\Rightarrow 3^{18} \equiv 2 \pmod{23}$

Remainder = 2

$(3^{18})^2 \equiv 2^2 \pmod{23}$

$\Rightarrow 3^{36} \equiv 4 \pmod{23}$

$\Rightarrow 3^{36} \cdot 3^3 \equiv 16 \pmod{23}$

$\Rightarrow 3^{39} \equiv 16 \pmod{23}$

$\Rightarrow 3^{39} \cdot 3 \equiv 48 \pmod{23}$

$\Rightarrow 3^{40} \equiv 2 \pmod{23}$

① Find the least residue mod 4 of $x = 1^5 + 2^5 + 3^5 + \cdots + 100^5$

② Find the ... mod 12 of the Sum $S = 1! + 2! + 3! + \cdots + 100!$

Linear Congruence:

A Congruence of the form $ax \equiv b \pmod{m}$ where x is an unknown integer, is called a linear Congruence in one variable.

Solution of a linear Congruence:

An integer x_0 is said to be solution of the congruence $ax \equiv b \pmod{m}$ if $m \mid ax_0 - b$ i.e $ax_0 \equiv b \pmod{m}$.

Incongruent Solution:

Let x_1, x_2 be two solutions of $ax \equiv b \pmod{m}$.

If $x_1 \not\equiv x_2 \pmod{m}$ then x_1 and x_2 are called incongruent solutions of $ax \equiv b \pmod{m}$. (Procedure to find incongruent sol^ns of $ax \equiv b \pmod{m}$)

Theorem: Let a, b and m integers with $m > 0$ and $(a, m) = d$.

i) if $d \nmid b$ then $ax \equiv b \pmod{m}$ has no solutions.

ii) if $d \mid b$, then $ax \equiv b \pmod{m}$ has exactly d-incongruent solutions modulo m and they are given by $x_0 = x_0 + t\left(\frac{m}{d}\right)$

where $t = 0, 1, 2, \ldots (d-1)$ and x_0 is a solution of $ax \equiv b \pmod{m}$

iii) if $(a, m) = 1$ then $ax \equiv b \pmod{m}$ has unique solution (incongruent)

~~Problem~~ Inverse modulo m:

Given an integer a with $(a, m) = 1$, a solution of $ax \equiv 1 \pmod{m}$ is called an inverse of "a" modulo "m"

(or)

If $ab \equiv 1 \pmod{m}$ then a, b are said to be inverses modulo m. Also b is called inverse of a and a is called inverse of b under modulo m.

$9 \cdot 7 \equiv 1 \pmod{31} \Rightarrow$ 7 is an inverse of 9 modulo 31.

(or) 9 is an inverse of 7 modulo 31.

Note: To solve any congruence of the form $ax \equiv b \pmod{m}$.

If $\bar{a}$ is an inverse of a modulo m then $a\bar{a} \equiv 1 \pmod{m}$

To solve, $ax \equiv b \pmod{m}$

multiplying both sides of this congruence by $\bar{a}$, we get

$$\bar{a}(ax) \equiv \bar{a}b \pmod{m}$$

$$\Rightarrow x \equiv (\bar{a}b) \pmod{m}$$

properly Let p be a prime. The positive integer a is its own inverse modulo p if and only if $a \equiv 1 \pmod{p}$ (or) $a \equiv -1 \pmod{p}$

proof case(1):

$\Leftarrow$: If $a \equiv 1 \pmod{p}$ (or) $a = -1 \pmod{p}$

$\Rightarrow a^2 \equiv 1 \pmod{p} \Rightarrow a \cdot a \equiv 1 \pmod{p} \Rightarrow$ a is its own inverse modulo p

$\Rightarrow p \mid a^2 - 1 \Rightarrow p \mid a$

case(2):

$\Rightarrow$: If a is its own inverse modulo p then $a^2 = a \cdot a \equiv 1 \pmod{p}$

$\Rightarrow p \mid a^2 - 1 \Rightarrow p \mid (a-1)(a+1)$

$\Rightarrow p \mid a-1$ (or) $p \mid a+1$ ($\because$ p is prime)

$\Rightarrow a \equiv 1 \pmod{p}$ (or) $a \equiv -1 \pmod{p}$

Note: An integer a has an inverse under modulo m $\Leftrightarrow (a, m) = 1$.

① Solve: $5x \equiv 2 \pmod{26}$.

Sol^n Here $a=5,\ b=2,\ m=26$.

$(a,m) = (5,26) = 1$ and $1/2$

$\Rightarrow 5x \equiv 2 \pmod{26}$ has unique (incongruent) solutions.

Since $5x \equiv 2 \pmod{26}$, Multiples of 26 are 26, 52, 78,

Note $0 \equiv -52 \pmod{26}$

$\Rightarrow 5x \equiv -50 \pmod{26}$

$\Rightarrow x \equiv -10 \bmod\left(\frac{26}{(5,26)}\right)$

$\Rightarrow x \equiv -10 \bmod(26)$

$\Rightarrow x \equiv 16 \bmod 26$

$ac \equiv bc \pmod{m}$

$a \equiv b \pmod{\frac{m}{(c,m)}}$

The solution is 16

Other solutions are ---- -36, (-10, 16, 42, 68, 94, 120, ----)

② Solve $5x \equiv 7 \pmod{2023}$

Sol^n Here $a=5,\ b=7,\ m=2023$

$(a,m) = (5,2023) = 1$ and $1/7$

$\therefore 5x \equiv 7 \pmod{2023}$ has unique (incongruent) solution.

Since $5x \equiv 7 \pmod{2023}$, Multiples of 2023 are 2023, 4046, ----

$0 \equiv 2023 \pmod{2023}$

$5x \equiv 2030 \pmod{2023}$

$\Rightarrow x \equiv 406 \bmod\left(\frac{2023}{(5,2023)}\right)$

$x \equiv 406 \bmod\left(\frac{2023}{1}\right)$

$\Rightarrow x \equiv 406 \bmod 2023$

$\therefore$ Solution is 406

"U

called inverse of a and a is called

(3) Solve: $3x+2 \equiv 0 \pmod 7$

Soln. $3x \equiv -2 \pmod 7$

$\Rightarrow 3x \equiv 5 \pmod 7$

Here $a=3,\ b=5,\ m=7$

$(a,m) = (3,7) = 1 \Rightarrow$ It has unique soln.

Multiples of 7 are $7, 14, 21, 28, \ldots$

$\because 0 \equiv 7 \pmod 7$

$\therefore 3x \equiv 12 \pmod 7$

$\Rightarrow x \equiv 4 \bmod \left(\frac{7}{(3,7)}\right) = 4 \bmod 7$

$\therefore$ Solution is $\underline{4}$

(4) $20x \equiv 19 \pmod{43}$

Soln Here $a=20,\ b=19$

$m=43.$

$(20,43) = 1,\ 1/43$

$\therefore$ It has unique soln.

Multiples of 43 are $43, 86, 129, 172, 215, 258, 301, \ldots$

$20x \equiv 19 \pmod{43}$

$0 \equiv 301 \pmod{43}$

$20x \equiv 320 \pmod{43}$

$\Rightarrow x \equiv 16 \bmod \left(\frac{43}{(20,43)}\right)$

$\Rightarrow x \equiv 16 \bmod 43$

$\therefore x_0 = 16$

Case (2) Solve: $10x \equiv 15 \pmod{35}$

Here $a=10,\ b=15,\ m=35$

$d = (a,m) = (10,35) = 5$ and $5/15$

$\therefore$ It has 5 incongruent solutions and are given by $x = x_0 + t\left(\frac{m}{d}\right)$, where $t = 0,1,2,\ldots d-1$

i.e $x = x_0 + t\left(\frac{35}{5}\right) = x_0 + 7t,\ t=0,1,2,3,4$

where x_0 is any one solution.

we have $10x \equiv 15 \pmod{35}$

$\Rightarrow 2x \equiv 3 \bmod \left(\frac{35}{(5,35)}\right)$

$\Rightarrow 2x \equiv 3 \bmod 7$

Here $x_0 = 5$ is one solution

$2x \equiv 3 \pmod 7$

$0 \equiv 7 \pmod 7$

$2x \equiv 10 \pmod 7$

$x \equiv 5 \pmod 7$

$\therefore x = x_0 + t\left(\frac{m}{d}\right) = x_0 + 7t,\ t=0,1,2,3,4$

$\therefore x = 5 + 7t,\ t=0,1,2,3,4$

$x = 5, 12, 19, 26, 33.$ are incongruent solutions

Note: If a is large in the congruence $ax \equiv b \pmod m$ we divide a by m so that a is congruence to its remainder then we multiply both sides by x and apply transitive relation and solve it.

Solve: $259x \equiv 5 \pmod{11}$

Soln

Here $a = 259$, $b = 5$, $m = 11$

$(a, m) = (259, 11) = 1 \Rightarrow$ It has unique incongruent solution.

$259x \equiv 5 \pmod{11}$

Here $a = 259 >> 11 = m$

$11 \overline{)259}\ (23$, 22, 39, 33, 6

$\therefore 259 \equiv 6 \pmod{11}$

$\Rightarrow 259x \equiv 6x \pmod{11}$ (

$\therefore 6x \equiv 5 \pmod{11}$ by transitive and symmetric)

$(6, 11) = 1$, $b = 5$

multiples of 11 are 11, 22, 33, ...

$0 \equiv -11 \pmod{11}$, $6x \equiv 5 \pmod{11}$

$\Rightarrow 6x \equiv -6 \pmod{11}$

$\Rightarrow x \equiv -1 \bmod \left(\frac{11}{(6,11)}\right)$

$\Rightarrow x \equiv -1 \pmod{11}$

$\Rightarrow x \equiv 10 \pmod{11}$

Solution is $x_0 = 10$

Solve: $20x \equiv 30 \pmod{4}$

$a = 20$, $b = 30$, $m = 4$ $(a, m) = (20, 4) = 4 = d$

Here $d \nmid b$ i.e $4 \nmid 30$

$\therefore$ It has no solution.

Find an inverse of 33 under modulo 91.

Soln Consider the linear congruence $33x \equiv 1 \pmod{91}$

$a = 33$, $b = 1$, $m = 91$

$(a, m) = (33, 91) = 1 \Rightarrow$ It has unique solu

we have $33x \equiv 1 \pmod{91}$, Multiples of 91 are 91, 182, ...

$0 \equiv -91 \pmod{91}$

$\Rightarrow \quad 33x \equiv -90 \pmod{91}$

$\Rightarrow \quad 11x \equiv -30 \mod\left(\frac{91}{(3,91)}\right) = -30 \mod 91$

$\Rightarrow \quad 11x \equiv -30 \pmod{91}$

Multiples of 91 are 91, 182, ----

$0 \equiv -91 \pmod{91}$

$+$

$11x \equiv -121 \pmod{91}$

$\Rightarrow x \equiv -11 \mod\left(\frac{91}{(11,91)}\right) = -11 \pmod{91}$

$\Rightarrow \quad x \equiv 80 \pmod{91}$

$\therefore \quad x_0 = 80.$

Any Inverse of 33 under modulo 91 is 80 $(\because 33 \times 80 \equiv 1 \bmod 91)$

1) Find the inverse pairs of $\mathbb{Z}_5 = \{0,1,2,3,4\}$

2) show that $a + x \equiv b \pmod{m}$ has unique solution.

3) prove that $ab \equiv 0 \pmod 6 \not\Rightarrow$ either $a \equiv 0 \pmod 6$ (or) $b \equiv 0 \pmod 6$

4) If $ab \equiv 0 \pmod p$ where p is a prime. Prove that either $a \equiv 0 \pmod p$ (or) $b \equiv 0 \pmod p$

List all integers in the range $1 \le x \le 100$ that satisfy $x \equiv 7 \pmod{17}$

If $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ be a polynomial of nth degree with integer coefficients and $a \equiv b \pmod m$ then prove that $f(a) \equiv f(b) \pmod m$.

Find the remainder when $x = 1^5 + 2^5 + 3^5 + \cdots + 100^5$ is divided by 4.

Soln Here $m = 4$

$1 \equiv 1 \pmod 4 \Rightarrow 1^5 \equiv 1 \pmod 4$

$2 \equiv 2 \pmod 4 \Rightarrow 2^5 \equiv 0 \pmod 4$

$3 \equiv -1 \pmod 4 \Rightarrow 3^5 \equiv (-1)^5 \pmod 4 \equiv -1 \pmod 4$

$4 \equiv 0 \pmod 4 \Rightarrow 4^5 \equiv 0 \pmod 4$

$5 \equiv 1 \pmod 4 \implies 5^5 \equiv 1 \pmod 4$

$6 \equiv 2 \pmod 4 \implies 6^5 \equiv 0 \pmod 4$

$7 \equiv 3 \pmod 4 \implies 7^5 \equiv 3^5 \pmod 4 \equiv (-1) \pmod 4$

$8 \equiv 0 \pmod 4 \implies 8^5 \equiv 0 \pmod 4$

$\vdots \qquad\qquad\qquad \vdots$

$99 \equiv (-1) \pmod 4 \implies 99^5 \equiv (-1) \pmod 4$

$100 \equiv 0 \pmod 4 \implies 100^5 \equiv 0 \pmod 4$

$\therefore 1^5 + 2^5 + 3^5 + 4^5 + \cdots + 100^5 \equiv (1+0+(-1)+0+1+0+(-1)+0+\cdots+(-1)+0) \pmod 4$

$\equiv 0 \pmod 4$

$\therefore$ Remainder is 0

Chinese Remainder theorem

If $m_1, m_2, \ldots m_k$ are positive integers such that $(m_i, m_j) = 1$ $\forall i \neq j$ and $M = m_1 m_2 \cdots m_k$ and $b_i \in \mathbb{Z}$ $(i = 1, 2, \ldots n)$ then the system of congruence

$$\left.\begin{aligned} x &\equiv b_1 \pmod{m_1} \\ x &\equiv b_2 \pmod{m_2} \\ &\vdots \\ x &\equiv b_k \pmod{m_k} \end{aligned}\right\} \text{ has exactly one}$$

solution $x_0 \pmod M$ where $x_0 = \frac{M}{m_1} y_1 b_1 + \frac{M}{m_2} y_2 b_2 + \cdots + \frac{M}{m_k} y_k b_k$

where the integers y_i are defined by

$$\left(\frac{M}{m_i}\right) y_i \equiv 1 \pmod{m_i} \quad \forall i = 1, 2, \ldots, k.$$

Problem ① Find the solution of the system $x \equiv 5 \pmod 4$
$x \equiv 3 \pmod 7$
$x \equiv 2 \pmod 9$ by using chinese remainder theorem.

Soln we have $m_1 = 4,\ m_2 = 7,\ m_3 = 9,\ b_1 = 5,\ b_2 = 3,\ b_3 = 2$.

$\therefore M = m_1 m_2 m_3 = 4 \times 7 \times 9 = 252$ (since $(m_i, m_j) = 1\ \forall\ i \neq j$)

$$x_0 = \frac{M}{m_1} y_1 b_1 + \frac{M}{m_2} y_2 b_2 + \frac{M}{m_3} y_3 b_3$$

$$= \frac{252}{4}(5y_1) + \frac{252}{7}(3y_2) + \frac{252}{9}(2y_3)$$

$$x_0 = 315y_1 + 108y_2 + 56y_3$$

where y_i's are given by $\frac{M}{m_i} y_i \equiv 1 \pmod{m_i}\ \forall\ i = 1, 2, 3$.

i.e. $\frac{252}{4} y_1 \equiv 1 \pmod 4$

$\Rightarrow 63y_1 \equiv 1 \pmod 4$

Since $(63, 4) = 1$, unique soln

Multiples of 4 are 4, 8, 12, 16, 20, 24, 28, 32, ...

$0 \equiv 20 \pmod 4$

$63y_1 \equiv 21 \pmod 4$

$\Rightarrow 3y_1 \equiv 1 \bmod \left(\frac{4}{(21,4)}\right)$

$\Rightarrow 3y_1 \equiv 1 \bmod 4$

Multiples of 4, 8, ...

$0 \equiv -4 \bmod 4$

$3y_1 \equiv -3 \bmod 4$

$y_1 \equiv -1 \bmod \left(\frac{4}{(3,4)}\right)$

$\Rightarrow y_1 \equiv -1 \pmod 4$

$y_1 = 3, -1 \pmod 4$

$\frac{252}{7} y_2 \equiv 1 \pmod 7$

$36y_2 \equiv 1 \pmod 7$

$(36, 7) = 1$

Multiples of 7 are 7, 14, 21, 28, 35, ...

$0 \equiv 35 \pmod 7$

$36y_2 \equiv 36 \pmod 7$

$\Rightarrow y_2 \equiv 1 \pmod 7$

$y_2 = 1 \pmod 7$

$\frac{252}{9} y_3 \equiv 1 \pmod 9$

$28y_3 \equiv 1 \pmod 9$

$(28, 9) = 1$

Multiples of 9 are 9, 18, 27, ...

$0 \equiv 27 \pmod 9$

$28y_3 \equiv 28 \pmod 9$

$\Rightarrow y_3 \equiv 1 \pmod 9$

$y_3 \equiv 1 \bmod 9$

$\therefore x_0 = 315(-1) + 108(1) + 56(1)$

$= 0 - 315 + 108 + 56 = 164 - 315$

$= -151$

$\therefore$ The solution is $x \equiv x_0 \pmod M$

$= -151 \pmod{252}$

$= 101 \pmod{252}$

Ans: 101

(2) Solve: the system of Congruence $x \equiv 5 \pmod{11}$, $x \equiv 15 \pmod{31}$, $x \equiv 14 \pmod{29}$ by using chinese remainder theorem.

Sol(n) Here $m_1 = 11$, $m_2 = 31$, $m_3 = 29$ are pairwise relatively prime positive integers to each other.

$\therefore M = m_1 m_2 m_3 = 11 \times 31 \times 29 = 9889$

Given $b_1 = 5$, $b_2 = 15$, $b_3 = 14$

$$x_0 = \frac{M}{m_1} y_1 b_1 + \frac{M}{m_2} y_2 b_2 + \frac{M}{m_3} y_3 b_3$$

$$= \frac{9889}{11} y_1 (5) + \frac{9889}{31} (15 y_2) + \frac{9889}{29} (14 y_3)$$

where y_i's are given by $\frac{M}{m_i} y_i \equiv 1 \pmod{m_i} \quad \forall i = 1, 2, 3$.

Now $\frac{9889}{11} y_1 \equiv 1 \pmod{11}$

$\Rightarrow 899 y_1 \equiv 1 \pmod{11}$

Since $899 > 7 \times 11$

$899 \equiv 8 \pmod{11}$

$\Rightarrow 899 y_1 \equiv 8 y_1 \pmod{11}$

$\therefore 8 y_1 \equiv 1 \pmod{11}$

$(8, 11) = 1$

Multiples of 11 are 11, 22, 33

$0 \equiv -33 \pmod{11}$

$8 y_1 \equiv -32 \pmod{11}$

$\Rightarrow y_1 \equiv -4 \pmod{11}$ $(\because (8, 11) = 1)$

$\boxed{y_1 \equiv -4 \pmod{11}}$

$\frac{9889}{31} y_2 \equiv 1 \pmod{31}$

$319 y_2 \equiv 1 \pmod{31}$

$\because 319 > 7 \times 31$

$319 \equiv 9 \pmod{31}$

$\Rightarrow 319 y_2 \equiv 9 y_2 \pmod{31}$

$\therefore 9 y_2 \equiv 1 \pmod{31}$

Multiples of 31 are 31, 62, ...

$0 \equiv 62 \pmod{31}$

$9 y_2 \equiv 63 \pmod{31}$

$\therefore \boxed{y_2 \equiv 7 \pmod{31}}$

$\frac{9889}{29} y_3 \equiv 1 \pmod{29}$

$341 y_3 \equiv 1 \pmod{29}$

$341 > 7 \times 29$

$341 \equiv (-7) \pmod{29}$

$341 \equiv 22 \pmod{29}$

$341 y_3 \equiv 22 y_3 \pmod{29}$

$\therefore 22 y_3 \equiv 1 \pmod{29}$

$(22, 29) = 1$

Multiples of 29 are 29, 58, 87, ...

$0 \equiv 87 \pmod{29}$

$22 y_3 \equiv 88 \pmod{29}$

$\Rightarrow y_3 \equiv 4 \pmod{29}$

$\boxed{y_2 \equiv 7 \pmod{31}}$

$\therefore x_0 = 899 \times 5 y_1 + 319 (15 y_2) + 341 (14 y_3)$

$= 899 \times 5 \times (-4) + 319 (15 \times 7) + 341 (14 \times 4)$

$= 34611$

Solution is $x_0 \pmod{M} = 34611 \pmod{9889} = 4944 \pmod{9889}$

) Solve the system: $x \equiv 1 \pmod 3$; $x \equiv 2 \pmod 5$, $x \equiv 3 \pmod 7$

Soln

$x \equiv 1 \pmod 3 \longrightarrow ① \Longrightarrow x = 3t+1 \longrightarrow ④$

$x \equiv 2 \pmod 5 \longrightarrow ⑤$

$x \equiv 3 \pmod 7 \longrightarrow ③$

put $x = 3t+1$ in ⑤, we have $3t+1 \equiv 2 \pmod 5$

$\Longrightarrow 3t \equiv 1 \pmod 5$ (multiply on both sides with 2)

$\Longrightarrow t \equiv 2 \pmod 5$ ($\because 3.2 \equiv 1 \pmod 5$)

$\therefore t = 2+5u$

put $t = 2+5u$ in ④, we get $x = 3(2+5u)+1 = 15u+7$

put $x = 15u+7$ in ③, we get $15u+7 \equiv 3 \pmod 7$

$\Longrightarrow 15u \equiv -4 \pmod 7$

$\Longrightarrow 15u \equiv 3 \pmod 7$

$\Longrightarrow 5u \equiv 1 \bmod \left(\frac{7}{(5,7)}\right)$

$\Longrightarrow 5u \equiv 1 \pmod 7$

$\Longrightarrow u \equiv 3 \pmod 7$

$\therefore u = 3+7v$

$\therefore x = 15(3+7v)+7 = 105v+52$

$\Longrightarrow x = 105v+52$

$\Longrightarrow x \equiv 52 \pmod{105}$

which is the simultaneous solution.

Lemma: If a and b are positive integers, then the least positive residue of 2^a-1 modulo 2^b-1 is 2^r-1 where r is the least positive residue of a modulo b (i.e $a \equiv r \pmod b$).

Proof: From the division algorithm $a = bq+r$ where r is the least positive residue of a modulo b i.e $a \equiv r \pmod b$ i.e $0 \le r < b$

we have $2^a-1 = 2^{bq+r}-1 = (2^b)^q \cdot 2^r - 1$

$= 2^r\left((2^b)^q - 1 + k\right) - 1$

$= 2^r\left((2^b)^q - 1\right) + (2^r - 1)$

$$\Rightarrow 2^a - 1 = 2^r\left(2^b-1\right)\cdot\left(\left(2^b\right)^{q-1} + \left(2^b\right)^{q-2} + \left(2^b\right)^{q-3} + \cdots + 1\right) + \left(2^r - 1\right)$$

$$\Rightarrow 2^a - 1 = \left(2^b - 1\right)q + \left(2^r - 1\right)$$

where $q = 2^r\left(\left(2^b\right)^{q-1} + \left(2^b\right)^{q-2} + \left(2^b\right)^{q-3} + \cdots + 1\right) \in \mathbb{Z}$

and $0 \le 2^r - 1 < 2^b - 1$ since $0 \le r < b$

i.e The remainder when $2^a - 1$ is divided by $2^b - 1$ is $2^r - 1$.
This is the least possitive residue of $2^a - 1$ modulo $(2^b - 1)$.

Lemma: If a and b are positive integers, then the greatest common divisor of $2^a - 1$ and $2^b - 1$ is $2^{(a,b)} - 1$.

Proof When we perform the Euclidean algorithm with $a = r_0$ and $b = r_1$ we obtain

$$r_0 = r_1 q_1 + r_2 \quad , \quad 0 \le r_2 < r_1$$

$$r_1 = r_2 q_2 + r_3 \quad 0 \le r_3 < r_2$$

$$\vdots \qquad \vdots$$

$$r_{n-3} = r_{n-2} q_{n-2} + r_{n-1} \quad , \quad 0 \le r_{n-1} < r_{n-2}$$

$$r_{n-2} = r_{n-1} \cdot q_{n-1}$$

where the least remainder is r_{n-1}

$$\therefore (a, b) = r_{n-1}.$$

using previous lemma, and the steps of the euclidean algorithm with $a = r_0$ and $b = r_1$, when we perform the euclidean algorithm on the pair $2^a - 1 = R_0$ and $2^b - 1 = R_1$,

we obtain $R_0 = R_1 Q_1 + R_2$ where $R_2 = 2^{r_2} - 1$

$R_1 = R_2 Q_2 + R_3$ where $R_3 = 2^{r_3} - 1$

$\vdots$

$R_{n-3} = R_{n-2} Q_{n-2} + R_{n-1}$ where $R_{n-1} = 2^{r_{n-1}} - 1$

$R_{n-2} = R_{n-1} Q_{n-1}$

Here the last non-zero remainder is $R_{n-1} = 2^{r_{n-1}} - 1$

$\Longrightarrow R_{n-1} = 2^{(a,b)} - 1$

$\therefore (R_0, R_1) = R_{n-1} = 2^{(a,b)} - 1$

i.e $(2^a - 1, 2^b - 1) = 2^{(a,b)} - 1$

Note. The positive integers $2^a - 1$ and $2^b - 1$ are relatively prime if and only if a and b are relatively prime.

Problem (4) A troop of 17 monkeys store their bananas in 11 piles of equal size, each containing more than 1 banana, with a twelfth pile of left over. When they divide the bananas into 17 equal groups, none remain. What is the smallest number of bananas they can have?

Soln Let x be the number of bananas

Given that $x \equiv 6 \pmod{11}$ and $x \equiv 0 \pmod{17}$

Here $m_1 = 11$, $m_2 = 17$ $b_1 = 6$, $b_2 = 0$

$(m_1, m_2) = (11, 17) = 1$.

CRT, $\therefore M = m_1 m_2 = 11 \times 17 = 187$.

$x \equiv x_0 \pmod{M} \equiv x_0 \pmod{187}$

where $x_0 = \frac{M}{m_1} y_1 b_1 + \frac{M}{m_2} y_2 b_2 = \frac{11\times17}{11} y_1(6) + \frac{11\times17}{17} y_2(0)$

$= 17 y_1(6) + 0$

$= 17 \times 6 y_1$

where y_1 is given by $\frac{M}{m_1} y_1 \equiv 1 \pmod{m_1}$

i.e $\frac{11\times17}{11} y_1 \equiv 1 \pmod{11}$

$\Rightarrow 17 y_1 \equiv 1 \pmod{11}$

$\Rightarrow 6 y_1 \equiv 1 \pmod{11}$

Multiples of 11 are 11, 22, 33, ...

$0 \equiv 11 \pmod{11}$

$6 y_1 \equiv 12 \pmod{11}$

$\Rightarrow y_1 \equiv 2 \bmod \left(\frac{11}{(6,11)}\right) \equiv 2 \pmod{11}$

$y_1 = 2$

$\therefore x_0 = 17 \times 6 \times 2 = 204$

$\therefore x \equiv x_0 \pmod{M} \equiv 204 \pmod{187} \equiv 17 \pmod{187}$

∴ The number of bananas takes the form $x = 17 + 187k$, where k is a non-negative integer.

Because each of the 11 piles contain more than 1 banana and 12th pile has 6 bananas, $\therefore x \geq 11(1) + 6 = 17$ i.e $x > 17$.

So, the smallest number of bananas is $17 + 187 = 204$

Multiplying both sides of this congruence by $\bar{\Delta}$, an inverse of Δ mod m (Δ modulo m)

we get $x \equiv \bar{\Delta}(de - bf) \pmod{m}$

~~Next~~, In a similar, way,

① × c $\Rightarrow$ $acx + bcy \equiv ce \pmod{m}$

② × a $\Rightarrow$ $acx + ady \equiv af \pmod{m}$

$$(bc - ad)y \equiv (ce - af) \pmod{m}$$

$$\Rightarrow (ad - bc)y \equiv (af - ce) \pmod{m}$$

$$\Rightarrow \Delta y \equiv (af - ce) \pmod{m}$$

multiplying both sides of this congruence by $\bar{\Delta}$, an inverse of Δ modulo m

we get $y \equiv \bar{\Delta}(af - ce) \pmod{m}$

when $x \equiv \bar{\Delta}(de - bf) \pmod{m}$ and $y \equiv \bar{\Delta}(af - ce) \pmod{m}$, then

we have $ax + by \equiv a(\bar{\Delta}(de - bf) \pmod{m}) + b(\bar{\Delta}(af - ce) \pmod{m})$

$$= \bar{\Delta}(ade - abf + abf - bce) \bmod m$$

$$\equiv \bar{\Delta}(ad - bc)e \pmod{m}$$

$$\equiv \bar{\Delta}\Delta e \pmod{m}$$

$$\equiv e \pmod{m}$$

and $cx + dy \equiv c\,\bar{\Delta}(de - bf) \pmod{m} + d(\bar{\Delta}(af - ce))$

$$\equiv \bar{\Delta}(cde - bcf + daf - dce) \pmod{m}$$

$$\equiv \bar{\Delta}(da - bc)f \pmod{m}$$

$$\equiv \bar{\Delta}\Delta f \pmod{m}$$

$$\equiv f \pmod{m}$$

ie we have shown that if (x, y) is a solution of the system of congruences then $x \equiv \bar{\Delta}(de - bf) \pmod{m}$, $y \equiv \bar{\Delta}(af - ce) \pmod{m}$

Defn. Let A and B be $n \times k$ matrices with integer entries, with $(i,j)^{th}$ entries a_{ij} and b_{ij} respectively. We say that A is congruent to B modulo m if $a_{ij} \equiv b_{ij} \pmod{m}$ for all pairs $(i;j)$ with $1 \le i \le n$ and $1 \le j \le k$. We write it as $A \equiv B \pmod{m}$ if A is congruent to B (modulo m).

eg. $\begin{bmatrix} 15 & 3 \\ 8 & 12 \end{bmatrix} \equiv \begin{bmatrix} 4 & 3 \\ 8 & 1 \end{bmatrix} \pmod{11}$

Proposition: If A and B are $n \times k$ matrices with $A \equiv B \pmod{m}$, C is an $k \times p$ matrix and D is $p \times n$ matrix, all with integer entries then $AC \equiv BC \pmod{m}$ and $DA \equiv DB \pmod{m}$.

Proof: Let the entries of A and B be a_{ij} and b_{ij} resply, for $1 \le i \le n$ and $1 \le j \le k$. and let the entries of C be c_{ij} for $1 \le i \le k$ and $1 \le j \le p$. The $(i,j)^{th}$ entries of AC and BC are $\sum_{t=1}^{k} a_{it} c_{tj}$ and $\sum_{t=1}^{k} b_{it} c_{tj}$ respectively.

Since $A \equiv B \pmod{m}$ we have $a_{it} \equiv b_{it} \pmod{m}$ $\forall$ i and t.

$$\Rightarrow a_{it} c_{tj} \equiv b_{it} c_{tj} \pmod{m}$$

$$\Rightarrow \sum_{t=1}^{k} a_{it} c_{tj} \equiv \sum_{t=1}^{k} b_{it} c_{tj} \pmod{m}$$

$$\Rightarrow AC \equiv BC \pmod{m}$$

Similarly, we can prove that $DA \equiv DB \pmod m$

Defn **Inverse of a matrix modulo m:**

If A and $\bar{A}$ are $n \times n$ matrices of integers and if $A\bar{A} \equiv \bar{A}A \equiv I \pmod m$, where $I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$ is the identity matrix of order n, then $\bar{A}$ is said to be an inverse of A modulo m.

Note: If $\bar{A}$ is an inverse of A modulo m and $B \equiv \bar{A} \pmod m$, then B is also an inverse of A modulo m.

Conversely, if B_1 and B_2 are both inverses of A modulo m then $B_1 \equiv B_2 \pmod m$.

Proof.

Case ① If $\bar{A}$ is an inverse of A then $A\bar{A} \equiv \bar{A}A \equiv I \pmod m$

and we have $B \equiv \bar{A} \pmod m$

$\Rightarrow AB \equiv A\bar{A} \pmod m \equiv I \pmod m$ ($\because$ From above proposition $BA \equiv \bar{A}A \pmod m \equiv I \pmod m$)

$\Rightarrow AB \equiv I \pmod m$

$\Rightarrow B$ is also an inverse of A modulo m

Case ② If B_1 and B_2 are both inverses of A then $AB_1 \equiv B_1A \equiv I \pmod m$

and $AB_2 \equiv B_2A \equiv I \pmod m$

Consider $AB_1 \equiv B_1A \equiv I \pmod m$

$\Rightarrow B_2(AB_1) \equiv B_2 \pmod m$

$\Rightarrow (B_2A)B_1 \equiv B_2 \pmod m$

$\Rightarrow B_1 \equiv B_2 \pmod m$ ($\because B_2A \equiv I \pmod m$)

eg The matrix $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ is an inverse of $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ modulo 5.

since $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 11 & 25 \\ 5 & 11 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \pmod 5$

and $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ 10 & 6 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \pmod 5$.

Method for finding inverses for 2×2 matrices

Proposition: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a matrix of integers, such that $\Delta \equiv \det A = ad - bc$ is relatively prime to the positive integer m. Then, the matrix $\bar{A} = \bar{\Delta}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ is an inverse of A (modulo m), where $\bar{\Delta}$ is the inverse of Δ modulo m.

Proof Consider $A\bar{A} \equiv \begin{bmatrix} a & b \\ c & d \end{bmatrix}\bar{\Delta}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \equiv \bar{\Delta}\begin{bmatrix} ad-bc & -ab+ab \\ dc-cd & ad-bc \end{bmatrix}$

$$\equiv \bar{\Delta}\begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

$$\equiv \bar{\Delta}\begin{bmatrix} \Delta & 0 \\ 0 & \Delta \end{bmatrix} \equiv \bar{\Delta}\Delta\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \equiv I \pmod m$$

and $\bar{A}A \equiv \bar{\Delta}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\equiv \bar{\Delta}\begin{bmatrix} ad-bc & bd-bd \\ -ac+ac & ad-bc \end{bmatrix} \equiv \bar{\Delta}\begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} \equiv \bar{\Delta}\begin{bmatrix} \Delta & 0 \\ 0 & \Delta \end{bmatrix}$$

$$\equiv \bar{\Delta}\Delta\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\equiv I \pmod m$$

where $\bar{\Delta}$ is an inverse of $\Delta \pmod m$, which exists because

Find the inverse of $A = \begin{bmatrix} 3 & 4 \\ 2 & 5 \end{bmatrix}$ modulo 13.

Soln Given $A = \begin{bmatrix} 3 & 4 \\ 2 & 5 \end{bmatrix}$ $\Delta = \det A = 3(5) - 2(4) = 15 - 8 = 7$

$m = 13$

since $(\Delta, m) = (7, 13) = 1$. $\bar{\Delta} = \bar{7} = 2$. since $7 \times 2 = 14 \equiv 1 \pmod{13}$

∴ Inverse of A modulo 13 exists.

and inverse of A modulo 13 is given by

$$\bar{A} \equiv \bar{\Delta} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{m}$$

$$= 2 \begin{bmatrix} 5 & -4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -4 & 6 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 9 & 6 \end{bmatrix} \pmod{13}$$

$$\therefore \bar{A} \equiv \begin{bmatrix} 10 & 5 \\ 9 & 6 \end{bmatrix} \pmod{13}$$

Defn **Adjoint of an $n \times n$ matrix**

The adjoint of an $n \times n$ matrix A is the $n \times n$ matrix with $(i,j)^{th}$ entry C_{ji}, where $C_{ji} = (-1)^{i+j} \cdot M_{ij}$. where M_{ij} is the determinant of the matrix obtained by deleting the i^{th} row and j^{th} column from A.

The adjoint of A is denoted by adj(A).

heorem: If A is an $n \times n$ matrix with $\det A \neq 0$, then $A(\text{adj}A) = (\det A) I$. where $\text{adj}A$ is the adjoint of A.

proposition: *

If A is an $n \times n$ matrix with integer entries and m is a positive integer such that $(\det A, m) = 1$, then the matrix $\bar{A} = \bar{\Delta}(\text{adj}A)$ is an inverse of A modulo m, where $\bar{\Delta}$ is the inverse of $\Delta = \det A$ modulo m.

roof

If $(\det A, m) = 1 \Rightarrow \det A \neq 0$.

$\therefore$ we have $A.\text{adj}A = \det A\,(\det A).I = \Delta I$

since $(\det A, m) = 1 \Rightarrow$ There is an inverse $\bar{\Delta}$ of $\Delta = \det A$ modulo m.

Hence $A(\bar{\Delta}\,\text{adj}A) \equiv (A.\text{adj}A)\,\bar{\Delta} \equiv \Delta\bar{\Delta} I \equiv I \pmod m$

and $\bar{\Delta}(\text{adj}A).A \equiv \bar{\Delta}(\text{adj}A.A) \equiv \bar{\Delta}(\Delta I) \equiv I \pmod m$

$\therefore$ $\bar{A} \equiv \bar{\Delta}(\text{adj}A)$ is an inverse of A modulo m.

oblem: Find the inverse of $A = \begin{bmatrix} 2 & 5 & 6 \\ 2 & 0 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ modulo 7.

soln Given $A = \begin{bmatrix} 2 & 5 & 6 \\ 2 & 0 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, $|A| = 2(0-4) - 5(6-2) + 6(4-0)$

$= -8 - 20 + 24$

$\Delta = -28 + 24 = -4 \equiv 3 \pmod 7$

$(\det A, 7) = (-3, 7) = 1$

$\therefore$ Inverse of Δ modulo 7 is 5 ie $\bar{\Delta} \equiv 5 \pmod 7$ ($\because 3 \times 5 \equiv 1 \pmod 7$)

$$\bar{A} = \bar{\Delta}(\text{adj}A) = 5 \begin{bmatrix} -4 & -4 & 4 \\ -3 & 0 & +1 \\ 10 & +8 & -10 \end{bmatrix}^T = 5 \begin{bmatrix} -4 & -3 & 10 \\ -4 & 0 & 8 \\ 4 & 1 & -10 \end{bmatrix} = \begin{bmatrix} -20 & -15 & 50 \\ -20 & 0 & 40 \\ 20 & 5 & -50 \end{bmatrix}$$

$\Rightarrow \bar{A} = \begin{bmatrix} -20 & -15 & 50 \\ -20 & 0 & 40 \\ 20 & 5 & -50 \end{bmatrix} \equiv \begin{bmatrix} 1 & 6 & 1 \\ 1 & 0 & 5 \\ 6 & 5 & 6 \end{bmatrix} \pmod 7$

Solution of the system $AX \equiv B \pmod m$

If $(\det A, m) = 1$. Then the solution x of the system $AX \equiv B \pmod m$ is given by

$$x \equiv \bar{A} B \pmod m$$

where $\bar{A}$ is an inverse of A modulo m and $\bar{A} = \bar{\Delta} \operatorname{adj} A \pmod m$

Problem: Solve the system of Congruences

$$2x_1 + 5x_2 + 6x_3 \equiv 3 \pmod 7$$
$$2x_1 + 2x_3 \equiv 4 \pmod 7$$
$$x_1 + 2x_2 + 3x_3 \equiv 1 \pmod 7$$

Sol: This is equivalent to the matrix Congruence

$$\begin{bmatrix} 2 & 5 & 6 \\ 2 & 0 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \equiv \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \pmod 7$$

It is of the form $AX \equiv B \pmod 7$

where $A = \begin{bmatrix} 2 & 5 & 6 \\ 2 & 0 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$ $m = 7$

The inverse of $A \pmod 7$ is $\bar{A} \equiv \begin{bmatrix} 1 & 6 & 1 \\ 1 & 0 & 5 \\ 6 & 5 & 6 \end{bmatrix} \pmod 7$.

∴ The solution of the system is

$$X \equiv \bar{A}B \pmod 7$$

$$\Rightarrow X \equiv \begin{bmatrix} 1 & 6 & 1 \\ 1 & 0 & 5 \\ 6 & 5 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \pmod 7$$

$$\equiv \begin{bmatrix} 3+24+1 \\ 3+5 \\ 18+20+6 \end{bmatrix} \bmod 7 \equiv \begin{bmatrix} 028 \\ 8 \\ 44 \end{bmatrix} \pmod 7$$

$$\equiv \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \pmod 7$$

∴ The solution is $X \equiv \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \pmod 7$

Hw

① Find the solutions of the following systems of linear congruences.

(a) $2x+3y \equiv 5 \pmod 7$
$x+5y \equiv 6 \pmod 7$

(b) $4x+y \equiv 2 \pmod 5$
$2x+3y \equiv 1 \pmod 5$

(c) $x+y \equiv 1 \pmod 7$
$x+z \equiv 2 \pmod 7$
$y+z \equiv 3 \pmod 7$

(d) $2x+y+z \equiv 1 \pmod 5$
$x+2y+z \equiv 1 \pmod 5$
$x+y+2z \equiv 1 \pmod 5$

② Find an inverse modulo 7 of each of the following matrices

a) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 5 \\ 1 & 4 & 6 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$